



Ministry of Industry and Trade



Hợp tác
Đức

DEUTSCHE ZUSAMMENARBEIT

Implemented by

giz Deutsche Gesellschaft
für Internationale
Zusammenarbeit (GIZ) GmbH

Economics Assessment of
**SOLAR PLUS STORAGE FOR
INDUSTRIAL & COMMERCIAL CONSUMERS
UNDER CURRENT TIME-OF-USE (TOU)
TARIFFS IN VIETNAM**

Imprint

Published by

Deutsche Gesellschaft für
Internationale Zusammenarbeit (GIZ) GmbH

Registered offices

Bonn and Eschborn, Germany

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Viet Nam (TEV)

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As at

November 2025

Photo credits

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GIZ is responsible for the content of this publication.

On behalf of

The German Federal Ministry for Economic
Cooperation and Development (BMZ)

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Abbreviations

ESCO	Energy Service Company
ESP	Energy Support Programme
EVN	Electricity of Vietnam
FIT	Feed-In-Tariff
GW	Gigawatt
HCMC	Ho Chi Minh City
IRR	Internal rate of return
kW	Kilowatt
kWh	Kilowatt-hour
MOIT	Ministry of Industry and Trade
MW	Megawatt
MWp	Megawatt peak
PDP8	Power Development Plan 8
PV	Photovoltaic
RTS	Rooftop Solar System
TCVN	Technical Standards of Vietnam
VND	Vietnamese dong (Đồng)
W	Watt
yr	Year



Executive summary

Rooftop solar (RTS) power, when paired with battery energy storage systems (BESS), is now economically competitive in Vietnam. Declining technology costs, coupled with Vietnam's existing time-of-use (TOU) tariff structures, are creating strong incentives for commercial and industrial (C&I) customers to adopt solar plus storage systems.

Evidence from Vietnam's current electricity demand curve shows that Vietnam's TOU tariffs already have a noticeable influence on the demand curve of C&I users, indicating that dynamic pricing has significant potential to further reshape the national load curve as adoption grows.

Across most regions of Vietnam, standalone rooftop solar PV systems currently achieve payback periods of **3–5 years**, making them an attractive investment for energy-intensive businesses. When battery storage is added, payback periods extend to approximately **6–9 years**, depending on regional conditions, whether debt financing is used, and depending on the specific tariff category applicable to the customer (commercial vs. industrial tariffs). Despite the longer return horizon, solar plus storage systems offer increased flexibility and resilience, potentially making them economically and strategically compelling for certain customers.

Modeling conducted for this analysis demonstrates that widespread deployment of solar-plus-storage among Vietnam's C&I sector would have a **major and noticeable impact** on the country's load curve. By reducing peak demand and shifting consumption, these systems can contribute meaningfully to national energy security, lower stress on the grid during high-load periods, and mitigate the need for costly peak-capacity investments.

This analysis covers the four customer types that constitute the largest categories of C&I users in Vietnam, namely steel, machinery, hotels, and food processing. Taken together, these four customer categories represent 50% of industrial power demand in the country.

However, rapid increases in distributed storage deployment introduce system-level challenges. In particular, large volumes of batteries charging simultaneously could create **new demand spikes**, shifting rather than eliminating stress points on the grid. To fully harness the benefits while minimizing unintended impacts, Vietnam may need targeted **policy and regulatory adjustments**, including improved planning frameworks, refined TOU structures, and new incentives or pricing structures that guide charging behavior in a grid-supportive manner.

Key findings of the report:

- Rooftop solar (RTS) is profitable in Vietnam today across all C&I users modeled, and can achieve payback times in the range of three(3) to five(5) years.
- The economics of adding BESS are less attractive overall, but can still make sense in certain applications, specifically to improve power reliability.
- Smaller BESS systems are more attractive on a financial basis than larger BESS systems. Also, a sensitivity analysis was conducted to examine higher TOU rates and found that introducing a “critical peak pricing” TOU tariff could help improve the financial attractiveness of BESS. However, such “critical peak prices” could have other undesirable consequences and may not be realistic in Vietnam at the present time. In order to mitigate these potential downsides, Vietnam may wish to consider applying “peak time rebates” instead, as have been introduced in certain US states, which offer a reduction on customers’ power bill for avoiding consuming during peak hours.
- On a consolidated basis (evaluating both RTS and BESS together), the payback times for RTS plus BESS for customers on industrial power tariffs are currently between 6 – 9 years; for hotels under commercial tariffs (which are currently higher than industry tariffs) the payback times are between 4-5 years depending on the location of the hotel, the solar resource in that location, and whether debt financing is used.
- Controlling the timing of BESS charging becomes increasingly critical as the total quantity of BESS grows. The analysis finds the simultaneous charging behavior, if allowed to proceed unchecked, could lead to new system demand peaks, and potentially even peak prices. Policies and pricing innovations are likely required to ensure smooth and more system-friendly BESS charging behavior.

Steel Sector Findings:

- Despite the fact that the total number of customers in this category is relatively small at 1.635 in total, each customer has a fairly large demand. As a result, the impacts of a few hundred customers adopting solar PV+BESS can be quite large.
- At a 50% RTS+BESS penetration rate, the reduction on system load reaches **8,300MW** during the daytime, which represents a **15% - 25% reduction** in peak electricity demand in Vietnam.

Food Processing Sector Findings:

- The overall impact on the power system demand curve is significant, particularly at higher levels of penetration: at an estimated 50% RTS+BESS penetration, the aggregate impact on system load is around **7,500MW** during the daytime.
- In terms of its contribution to mitigating peak electricity demand, the adoption of RTS+BESS in the food processing sector among roughly 50% of the sector’s customers would trigger a **15% - 25% reduction** in peak electricity demand.

Hotel Sector Findings:

- The overall impact on the power system demand curve is relatively small, even at higher levels of penetration: at an estimated 50% RTS+BESS penetration, the aggregate impact on system load is only around **650MW** during the daytime.
- In terms of its contribution to mitigating peak electricity demand, the adoption of RTS+BESS in the hotel sector among roughly 50% of the sector's customers would trigger a **1.5% - 2% reduction** in peak electricity demand.

Machinery Sector Findings:

- The overall impact on the power system demand curve is significant particularly at higher levels of penetration: at an estimated 50% RTS+BESS penetration, the aggregate impact on system load is up to **6.700MW** during the daytime.
- In terms of its contribution to mitigating peak electricity demand, the adoption of RTS + BESS in the machinery sector among roughly 50% of the sector's customers would trigger an **11% - 20% reduction** in peak electricity demand.

Overall, depending on the level of uptake that occurs in the C&I sector in the years ahead (which depends on policy, grid reliability, as well as potential changes to time-of-use tariffs), solar plus BESS systems can significantly reshape Vietnam power's demand curve. As such, solar plus BESS can play a vital role in supporting the grid, and mitigating demand peaks.

However, in order to avoid unintended consequences such as sudden demand peaks caused by simultaneous BESS charging, additional policy and regulatory interventions are likely needed. This could include further changes to the specific timing of TOU tariffs, or other policy interventions such as mandates, or BESS-specific requirements.

Taken together, rooftop solar combined with battery storage represents a powerful tool for Vietnam's evolving energy landscape – offering cost savings, enhanced reliability, and greater power system flexibility – provided that deployment is accompanied by appropriate policy and grid-management strategies.

Further analysis is needed to evaluate the impacts of the new two-part electricity tariff in Vietnam to better assess the impacts of this new tariff structure on the attractiveness of solar plus BESS.

1. Introduction

This analysis provides an overview of time-of-use (TOU) tariffs and their impacts on the business case for investments in customer-sited solar plus storage installations in Vietnam, as well as on the shape of the load curve. Different TOU tariff structures can incentivize customers to adjust their consumption from the grid, as well as the timing of their charging/discharging, providing valuable services to the grid.

In practice, this means that the battery is charged more during periods of low electricity prices. During expensive tariff periods, the battery storage system is then discharged to help cover demand.

TOU tariffs incentivize consumers to adjust their power demand during periods when rates are lower, which often correspond with periods of low demand, or of high supply, as in the case of highly windy or sunny days. While individual customer adjustments of the demand profile have a limited impact, when hundreds or even thousands of customers start to adjust their demand to TOU tariffs, the impact on the overall system load curve can be substantial.

Evidence from markets such as California, Texas, and Australia, among others, show that battery energy storage systems (BESS) are already starting to have a clear and measurable impact on the system load curve. With adjustments to Vietnam's TOU tariffs and a supportive enabling environment for BESS projects, Vietnam can achieve a number of related policy objectives including achieving a more reliable power system, reducing stress on the grid during peak hours, and reducing the need for costly system upgrades. This analysis is designed to look more closely at the adoption of rooftop solar (RTS) and BESS among commercial and industrial customers in Vietnam.

1.1. Vietnam Context

Commercial and industrial customers in Vietnam are increasingly expressing interest in adopting onsite battery energy storage systems (BESS). In most cases, such systems are coupled with customer-sited solar installations, enabling the user to reduce purchases from the grid by using solar power directly (direct self-consumption) while supplementing this ability by storing surplus power in onsite storage systems.

As can be seen below, the nationally approved average electricity price has increased from VND 1.622/kWh in 2015 to VND 2.204/kWh in 2025.

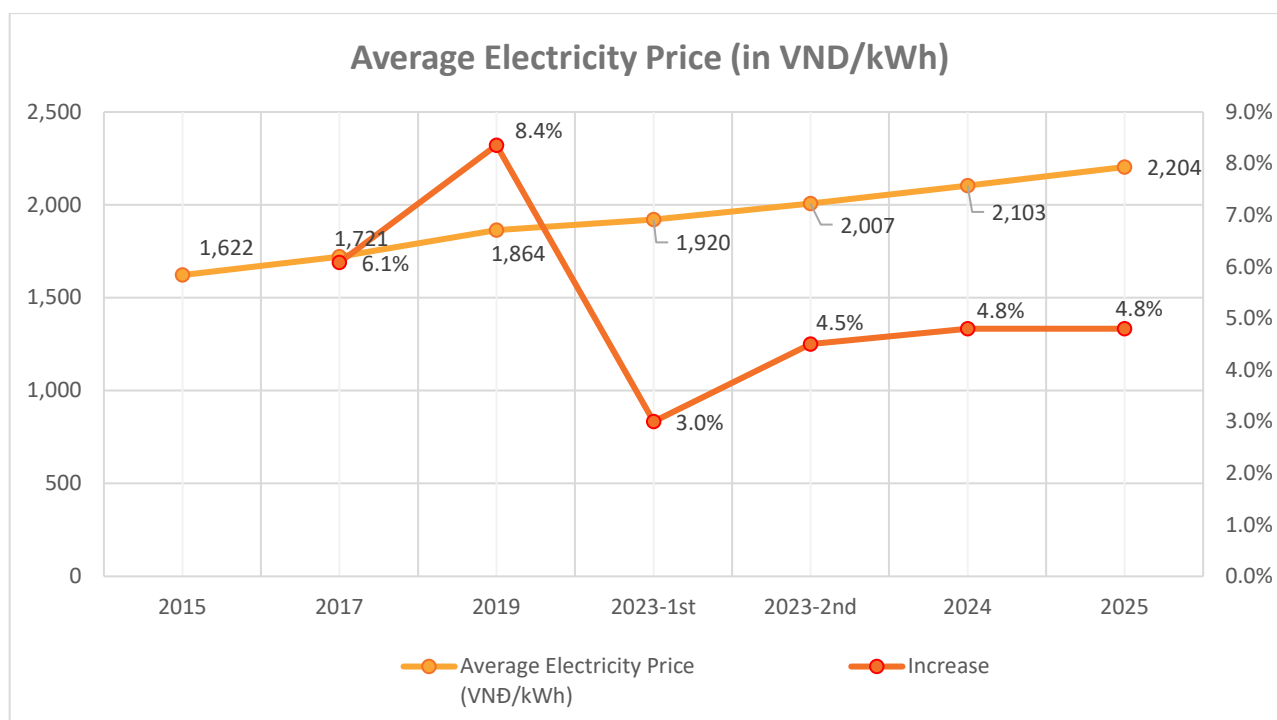


Figure 1. Electricity price increases in Vietnam (2015 – 2025)

More important the increase in average electricity prices is that Vietnam has relatively advanced TOU tariffs, with a significant spread between “off-peak” and “on-peak” electricity tariffs, particularly for commercial customers. **The upper end of the range of peak tariffs for commercial customers surpasses VND 5.000/kWh (USD \$0.19/kWh), which represents a tariff that is 3x higher than the “off-peak” tariffs applicable.**

The table below provides an overview of the tariffs according to Decision No. [1279/QĐ-BCT](#) dated 09 May 2025 on electricity prices. Since the focus of the analysis is on commercial and industrial customers, the assumed voltage level for all customers was between 22kV – 110kV.

No.	Customer Types	Rates (VND/kWh)		
		Regular hours	Off-peak hours	Peak hours
1	Industry/Production			
1.1	Voltage level from 110 kV and above	1,811	1,146	3,266
1.2	Voltage level from 22 kV to less than 110kV	1,833	1,190	3,398
1.3	Voltage level from 6 kV to less than 22kV	1,899	1,234	3,508
1.4	Voltage level less than 6kV	1,987	1,300	3,640
2	Commerce			
2.1	Voltage level from 22 kV and above	2,887	1,609	5,025
2.2	Voltage level from 6 kV to less than 22kV	3,108	1,829	5,202
2.3	Voltage level less than 6kV	3,152	1,918	5,422

Table 1. Summary of national TOU Tariffs (H1:2025)

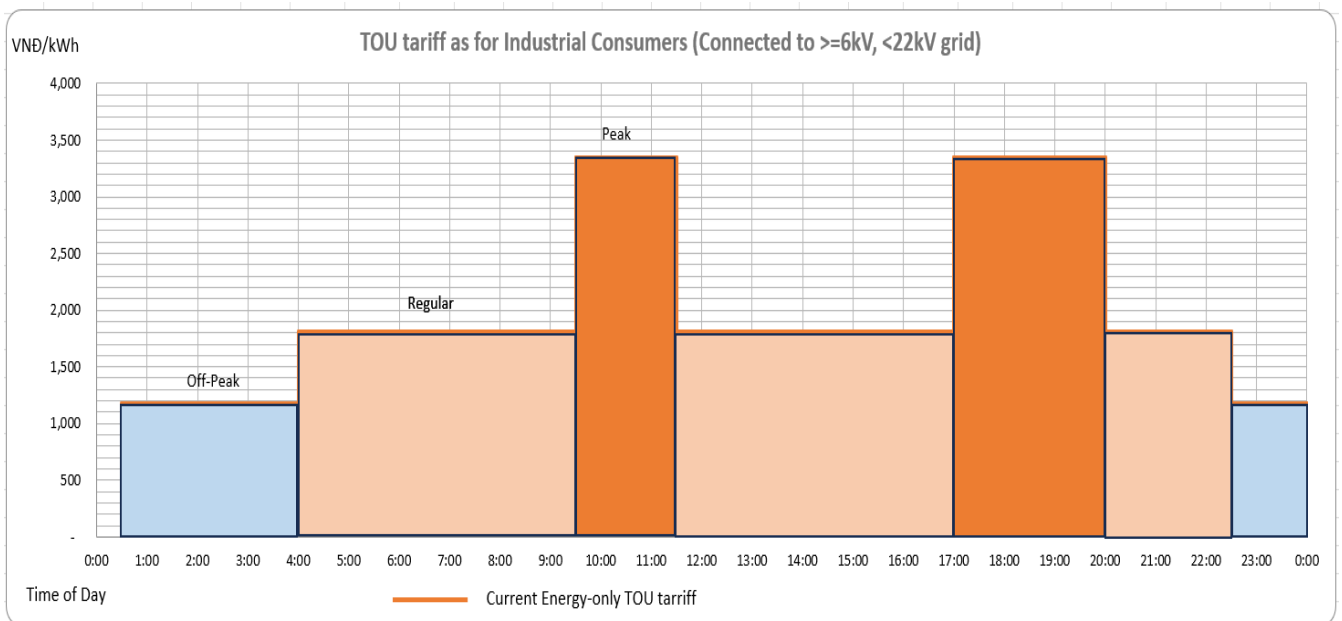


Figure 2. Current TOU Tariffs in Vietnam (H1:2025)

This relatively high spread between “on-peak” and off-peak tariffs creates an incentive to shift more electricity consumption off-peak, and thereby help reduce system peaks. Before the rise of BESS systems, however, it was often more difficult for C&I customers to reduce their demand, or to shift it around during the day to avoid the peak tariff rates. With BESS, this possibility has now become readily available to commercial and industrial customers, not only in Vietnam, but also worldwide.

At the same time, the steady increase in electricity prices combined with a growing desire for cleaner and more renewable electricity supply in their supply chains is leading a growing number of commercial and industrial customers in Vietnam to explore investments in onsite RTS+BESS projects. In some cases, companies with existing solar PV projects are looking to add storage to their existing solar PV systems, while some companies are considering entirely new investments in combined RTS+BESS projects.

1.2. International Context

The rapid reduction in BESS system costs has made BESS systems increasingly cost-competitive with retail electricity prices in many parts of the world. Even in jurisdictions with lower insolation levels such as Germany, onsite solar plus storage systems can help users reduce their overall energy costs.

The combined reductions of solar and BESS costs have transformed the business case for many commercial and industrial power users. This section is broken into two parts: the first part focuses on the latest cost trends for BESS and solar PV projects; the second part looks at a range of TOU tariffs from different markets around the world to provide comparisons for decision-makers in Vietnam.

1.2.1. Cost Trends for Solar and Storage Systems

The Levelized Costs Of Energy (LCOE) of solar plus storage have fallen 22% between Q1: 2024 and Q1:2025 alone, underscoring the rapid cost declines that are occurring worldwide¹. In fact, combining solar and storage is now cheaper than new coal (USD \$118/MWh) and nuclear (USD \$182/MWh).

Battery prices saw their biggest annual drop since 2017, reaching a price just under €100/kWh (converted from US\$) for the first time in 2024. Factors driving the decline include cell manufacturing overcapacity, economies of scale, low metal and component prices, adoption of lower-cost lithium-iron-phosphate (LFP) batteries, and a slowdown in electric vehicle sales growth². The figure below represents a global average, with prices varying widely across different countries and application areas.

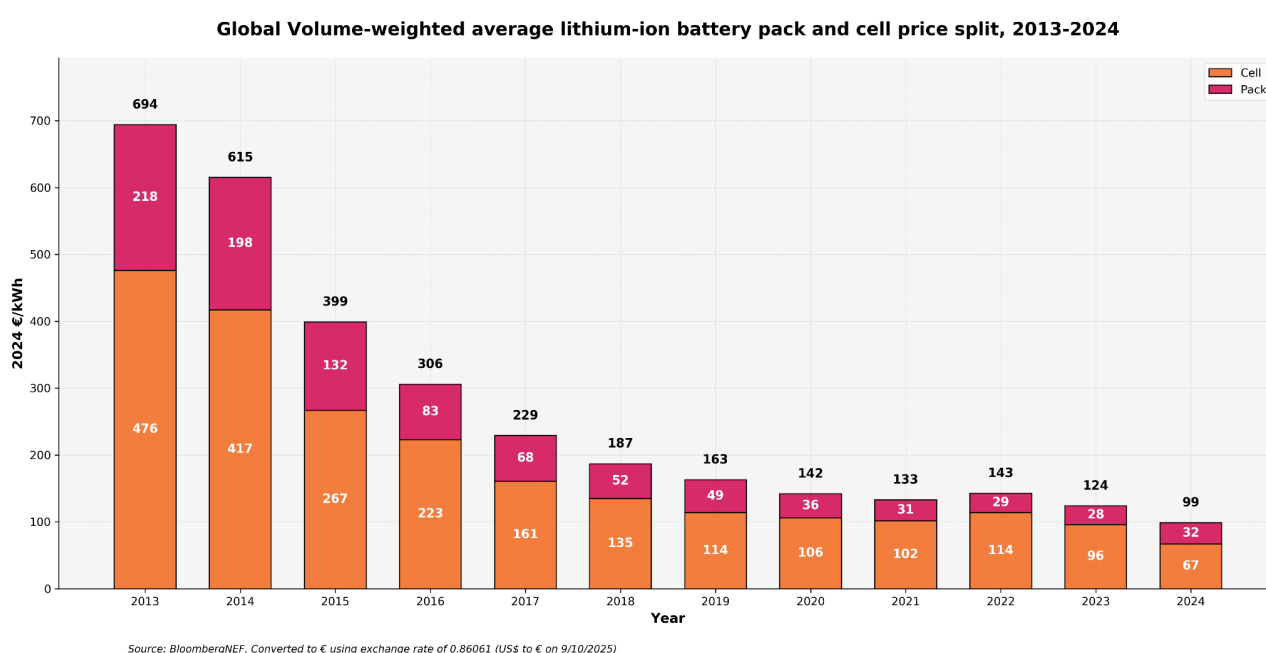


Figure 3. Decline in lithium-ion battery pack prices (2013-2024)

Between 2000 and 2024, global solar photovoltaic (PV) panel prices declined dramatically from around US\$ 6.3 per watt to US\$ 0.25 per watt (adjusted for inflation), representing a cost reduction of approximately 96%. This sustained downward trend reflects major advances in manufacturing efficiency, scale economies—particularly driven by Chinese production—and improvements in cell technology. After steep declines in the 2000s and early 2010s, prices stabilized at low levels post-2018, signaling a maturing market with incremental innovation. Between 2023 and 2024 alone, panel prices fell by roughly 20%, underscoring ongoing competitive pressure and technological refinement in the global solar supply chain.

¹ Ember (2025)

<https://ember-energy.org/app/uploads/2025/06/Ember-24-Hour-Solar-Electricity-June-2025-6.pdf>

² Bloomberg (2024)

<https://about.bnef.com/insights/commodities/lithium-ion-battery-pack-prices-see-largest-drop-since-2017-falling-to-115-per-kilowatt-hour-bloombergnef/>

Solar (photovoltaic) panel prices in US\$/W 2000-2024, adjusted for inflation

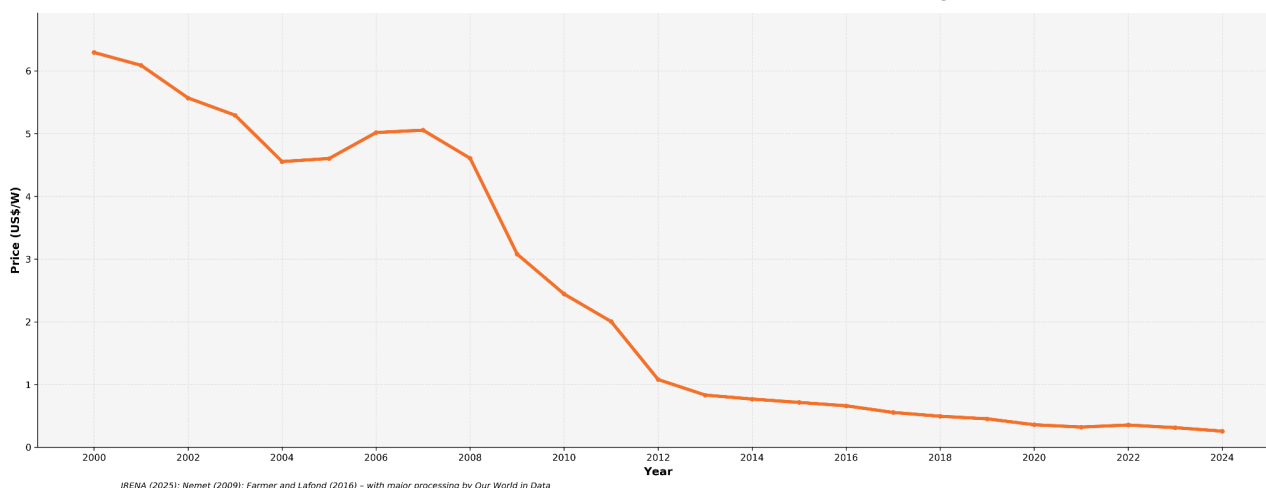


Figure 4. Decline in PV panel prices (2000 - 2024)

1.3. Global Overview of TOU Tariffs

This section provides a concise overview of TOU tariffs in select jurisdictions worldwide. The aim of the section is to provide insight into how other jurisdictions around the world (and other power utilities) are structuring their TOU tariffs, and to give a sense of how large the spread is between the peak- and off-peak tariffs. The size of this spread (combined with the absolute level of the electricity tariffs) is one of the main factors determining how attractive investments in BESS system are in a given jurisdiction.

1.3.1. California

In 2024, California generated a total of 278,338 GWh of electricity.³

Time-of-Use (ToU) tariffs are a core feature of California's electricity market, and have been made mandatory for businesses operating in the state. In many other jurisdictions in the world, TOU tariffs remain optional. In California, TOU tariffs are also strongly encouraged for residential users and are aimed at promoting greater demand-side flexibility. The state's three major investor-owned utilities (IOUs)—Pacific Gas and Electric (PG&E), Southern California Edison (SCE), and San Diego Gas & Electric (SDG&E)—each offer distinct ToU plans.

Across these utilities, peak pricing typically applies between 4:00 p.m. and 9:00 p.m., when demand is highest, while off-peak hours outside this window offer reduced rates.⁴ Seasonal

³ California Energy Commission (2025)

<https://www.energy.ca.gov/data-reports/energy-almanac/california-electricity-data/2024-total-system-electric-generation>

⁴ PG&E (2025).

Time of Use Rate Plans, <https://www.pge.com/en/account/rate-plans/time-of-use-rate-plans.html>; see also SCE (2025).

Time of Use Plans, <https://www.sce.com/save-money/rates-financing/residential-rate-plans/time-of-use-plans>. Liberty (2025). Time of Use Tariffs,

variations are also common, with lower overall prices during winter months and, in some cases, off-peak pricing extended throughout weekends and holidays, providing additional flexibility and cost-saving opportunities for consumers. At the upper end of the tariff range, some utility TOU tariffs reach nearly EUR 0.50/kWh (VND 15,300/kWh), which makes it very attractive for customers to self-supply with solar and add BESS systems onsite.

California has over 49 GW of installed solar capacity as of Q2 2025⁵. As of October 2025, there are 229 utility-scale battery storage projects in California, with total operating capacity at 14.064 MW.⁶

California - Liberty Commercial Time-of-Use Pricing

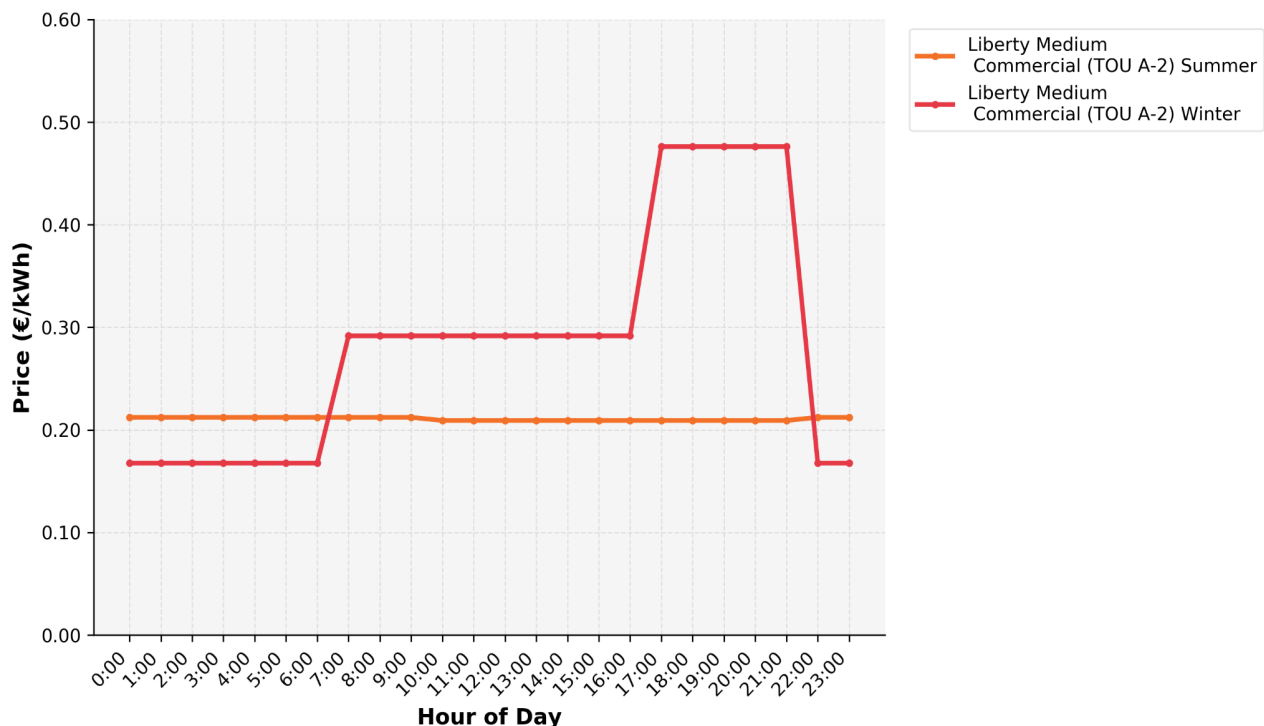


Figure 5. TOU pricing in California

1.3.2. China

In 2024, China generated approximately 10,073 TWh of electricity⁷, with clean energy sources contributing around 38% of total generation, including 18% from wind and solar power⁸.

<https://california.libertyutilities.com/verdi/residential/smart-energy-use/whats-the-tou-rate.html>

⁵ SEIA (2025)

<https://seia.org/state-solar-policy/california-solar/#:~:text=California%20has%20the%20largest%20solar,than%2031%20percent%20of%20California's>

⁶ Cleanview (2025)

<https://cleanview.co/battery-storage-projects/california>

⁷ Our World in Data (2025)

<https://ourworldindata.org/grapher/electricity-generation>

⁸ Ember (2025)

<https://ember-energy.org/countries-and-regions/china/>

TOU pricing has been gradually introduced in China since the 1980s, forming an integral part of China's broader energy market reform. **The National Development and Reform Commission (NDRC) continues to promote the expansion and optimization of TOU schemes to enhance demand-side flexibility and system efficiency.** Regional differentiation remains a key feature of China's TOU framework—some provinces implement seasonal variations (e.g., wet/dry or winter/summer periods), while others tailor pricing structures to local grid and resource conditions.

China's scale of renewable deployment is unparalleled. With the addition of 210 GW of solar power in the first half of 2025, China's total installed photovoltaic capacity has already surpassed 1.1 TW.⁹ Complementing this, the country has added over 100 GW of installed battery storage capacity as of the first half of 2025¹⁰. China is on track to have at least 2.461 GW of renewable electricity capacity installed by 2030, doubling the 2022 figure, with solar capacity nearly tripling over this time period.

China - Different provinces Commercial Time-of-Use Pricing

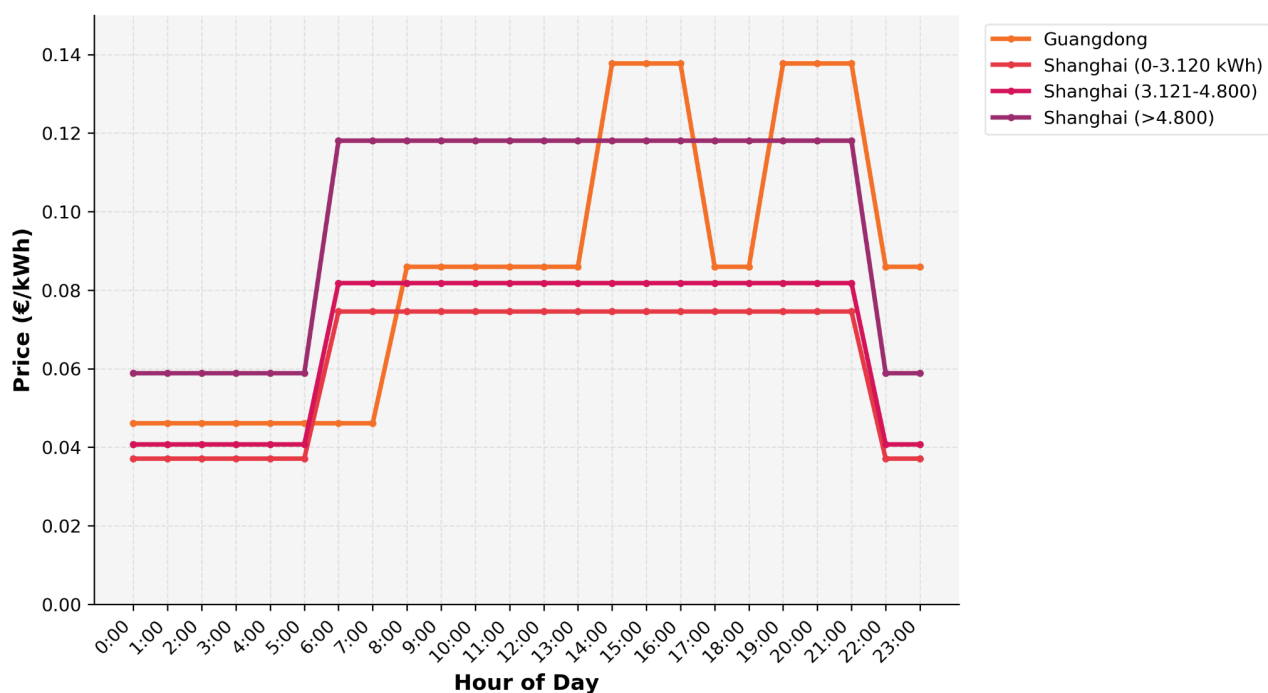


Figure 6. TOU pricing in China¹¹

⁹ IWR Online (2025)

<https://www.renewable-energy-industry.com/news/world/article-7018-china-becomes-the-first-country-in-the-world-to-surpass-1000-gw-of-solar-power-capacity>

¹⁰ IWR Online (2025)

<https://www.renewable-energy-industry.com/news/world/article-7027-chinas-energy-storage-market-booms-in-2025-battery-storage-capacity-exceeds-100-gw-for-the-first-time>

¹¹ Sources : Guo, J., et al. (2022). MDPI Journal, Processes 2022, 10, 832. <https://doi.org/10.3390/pr10050832>. https://www.researchgate.net/figure/ToU-pricing-in-Guangdong-China_tbl2_360166073; and Xiao, T., and Zhou, Q., (May 19 2025). China's industrial power rates 2025: A guide for investors, China Briefing: <https://www.china-briefing.com/news/chinas-industrial-power-rates-category-electricity-usage-region-classification/>

1.3.3. Australia

In 2024, Australia generated approximately 283,920 GWh of electricity, with renewable sources contributing around 102,403 GWh.¹²

TOU tariffs are not mandatory, but are available to customers across all states, offering flexibility between dynamic pricing structures and flat-rate plans. Despite being optional, TOU tariffs are adopted at relatively high rates, supported by a wide range of offerings from utilities and competitive retailers and supported by high levels of smart meter adoption. Depending on the province, TOU tariffs use off-peak and on-peak hours, and some provinces (or individual utilities) offer “shoulder” tariffs in between those two periods of the day.

This adoption is underpinned by Australia’s rapid growth in distributed energy resources. As of 30 June 2025, there are over 4.16 million PV installations in Australia, with a combined capacity of over 41.8 gigawatts.¹³ An expanding battery energy storage system (BESS) fleet with 13,417 MWh of installed capacity on the main grid complements this growth.¹⁴ Together, these developments enhance customer participation in the energy market and provide greater flexibility to the power system as a whole.

Victoria, Australia - Variable provider Commercial Time-of-Use Pricing

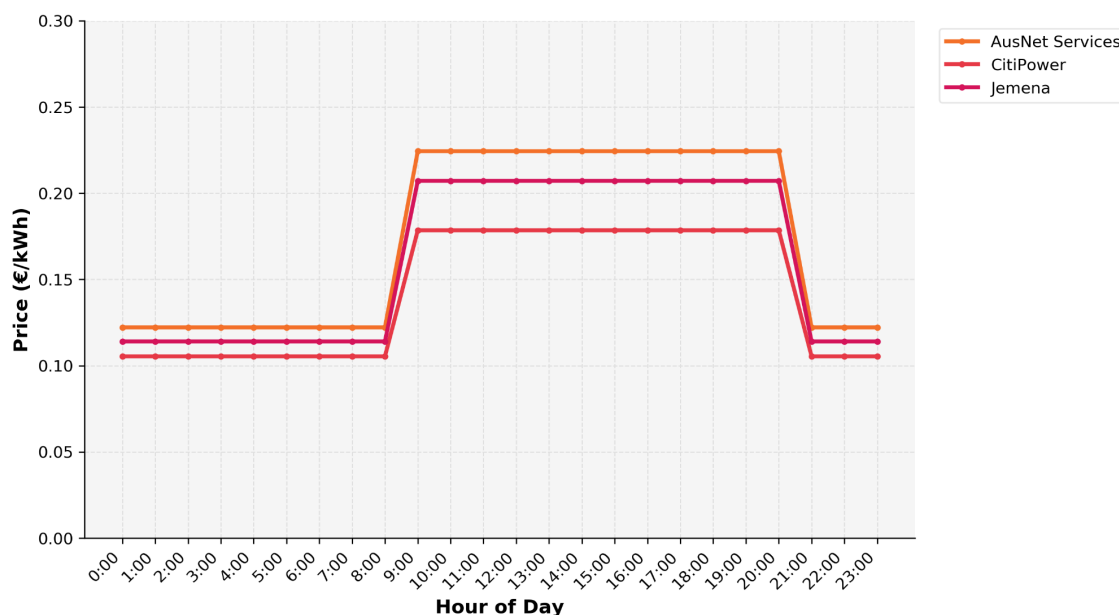


Figure 7. TOU pricing in Victoria, Australia

¹² Australian Department of Climate Change, Energy, the Environment and Water (2025)
<https://www.energy.gov.au/publications/australian-energy-statistics-table-o-electricity-generation-fuel-type-2023-24-and-2024>

¹³ Australian PV Institute (2025)
<https://pv-map.apvi.org.au/analyses>

¹⁴ Renew Economy (2025)
https://reneweconomy.com.au/graph-of-the-day-the-stunning-rise-in-big-battery-activity-on-australias-main-grids/#:~:text=Key%20Takeaways%20*%20Battery%20discharge%20rates%20have,to%2030%25%20of%20the%20evening%20peak%20supply.

1.3.4. Germany

In 2024, Germany generated 431.7 TWh of electricity, with renewable sources accounting for approximately 59% (254.9 TWh) of total production.¹⁵

The Energy Industry Act, amended in summer 2021, mandates that all energy suppliers offer dynamic electricity tariffs starting in 2025, enabling consumers to benefit from price signals that reflect real-time market conditions. Although not all suppliers have yet implemented these tariffs, the reform represents a significant milestone toward a more flexible and transparent energy market.

Further momentum came with the Act to Restart the Digitalization of the Energy Transition, passed in May 2023, which aims to accelerate smart meter deployment across the country. Smart meter adoption has lagged that of its neighbors like Sweden and Denmark, and the current penetration rate for smart meters is estimated at just over [2%](#), making Germany last across all EU countries in terms of smart meter adoption. Most TOU tariffs in Germany are now linked to hourly wholesale market prices, and since October 2025, some utilities like Vattenfall offer prices that are updated every 15 minutes, providing consumers with a high degree of exposure to real-time market dynamics. When linked to smart devices and algorithms that can trigger both upward and downward adjustments in demand, such exposure to market prices can also provide a high degree of responsiveness while contributing to reducing power demand peaks and reducing congestion within the system.

Supporting this digital and renewable expansion, Germany's installed solar capacity has now reached over 110 GW (as of [September 2025](#)), complemented by around 1.8 million battery storage systems with a combined capacity of 22.9 GWh ([September 2025](#)), significantly enhancing grid flexibility.

¹⁵ Bundesnetzagentur (2025)

https://www.bundesnetzagentur.de/SharedDocs/Pressemitteilungen/EN/2025/20250103_SMARD.html

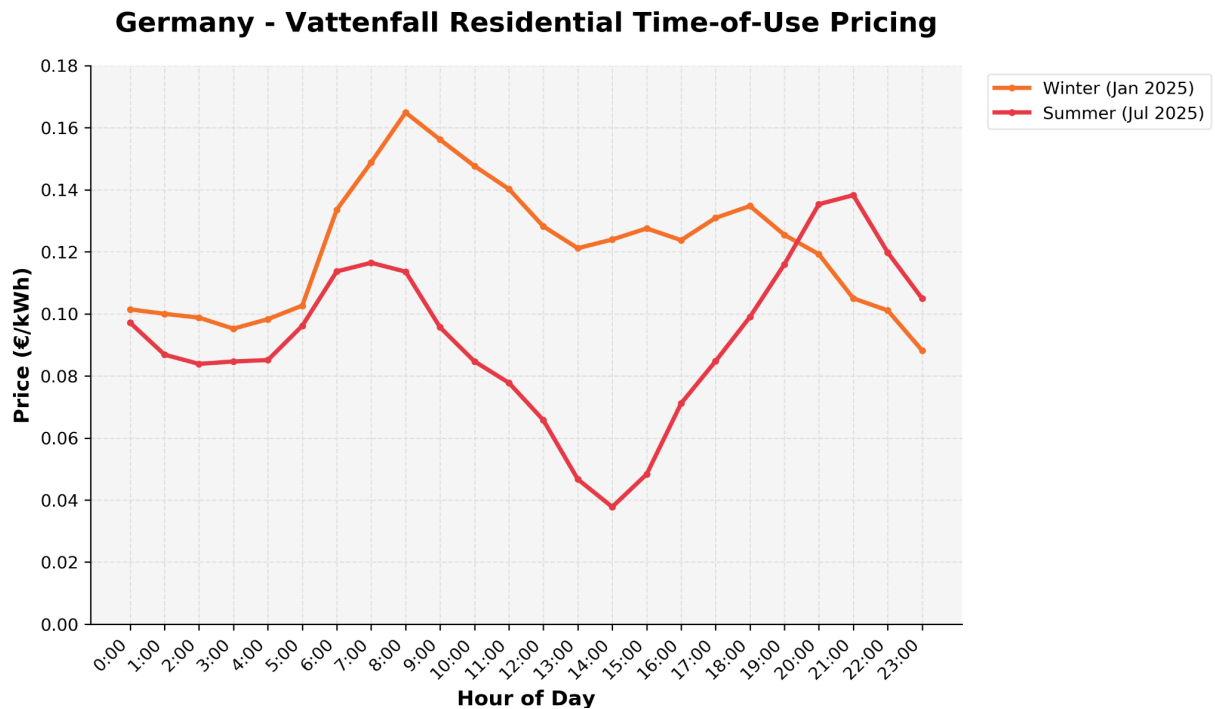


Figure 8. TOU pricing in Germany (Vattenfall)

1.3.5. UK

Electricity generated in the UK in 2024 amounted to a total of 285 TWh, with renewable generation at a record 144.7 TWh, accounting for 50.8% of UK total generation¹⁶.

In the United Kingdom, TOU tariffs are not mandatory for electricity suppliers, and their adoption remains relatively limited. The concept dates back to the 1970s with the introduction of the Economy 7 tariff, which offered reduced rates during seven off-peak nighttime hours. However, this early initiative did not lead to widespread uptake of dynamic pricing structures and customer-side demand response remained limited due to the lack of flexible appliances and the absence of smart meters and devices.

In recent years, the rollout of smart meters and the growing adoption of electric vehicles (EVs) have renewed consumer interest in TOU tariffs, enabling more flexible and efficient energy use. Adoption of TOU tariff plans increased by 75% in 2024, reflecting a clear upward trend.

Several suppliers now offer innovative options—most notably Octopus Energy’s “Agile” plan, which features half-hourly price updates based on wholesale market fluctuation. Other plans appeal particularly to EV users and tech-enabled households seeking to optimize charging and consumption patterns.

¹⁶ UK Department for Energy Security & Net Zero (2025)
https://assets.publishing.service.gov.uk/media/67e4f5d855239fa04d412067/Energy_Trends_March_2025.pdf

The UK now reports 14 GW of solar PV capacity¹⁷, with an additional 6.8GW/10.5GWh of operational battery storage¹⁸ as of September 2025, to complement the solar and wind power generated. As can be seen in the chart below showing Octopus Energy's residential TOU tariffs, the spread between on-peak and off-peak tariffs in early 2025 was nearly 6x, and surpassed EUR 1/kWh (VND 30.600/kWh) during critical peak periods on winter evenings. Since the TOU tariffs are linked to wholesale market prices, the actual tariff that customers who choose these "agile" tariffs have to pay cannot be known in advance: it fluctuates continuously according to supply and demand, and can even reach higher levels during periods of extreme scarcity. Such high pricing, combined with significant spreads between the highest and the lowest tariffs paid, currently make investments in BESS highly attractive in the UK market.

East Midlands, UK - Octopus Energy Residential Time-of-Use Pricing

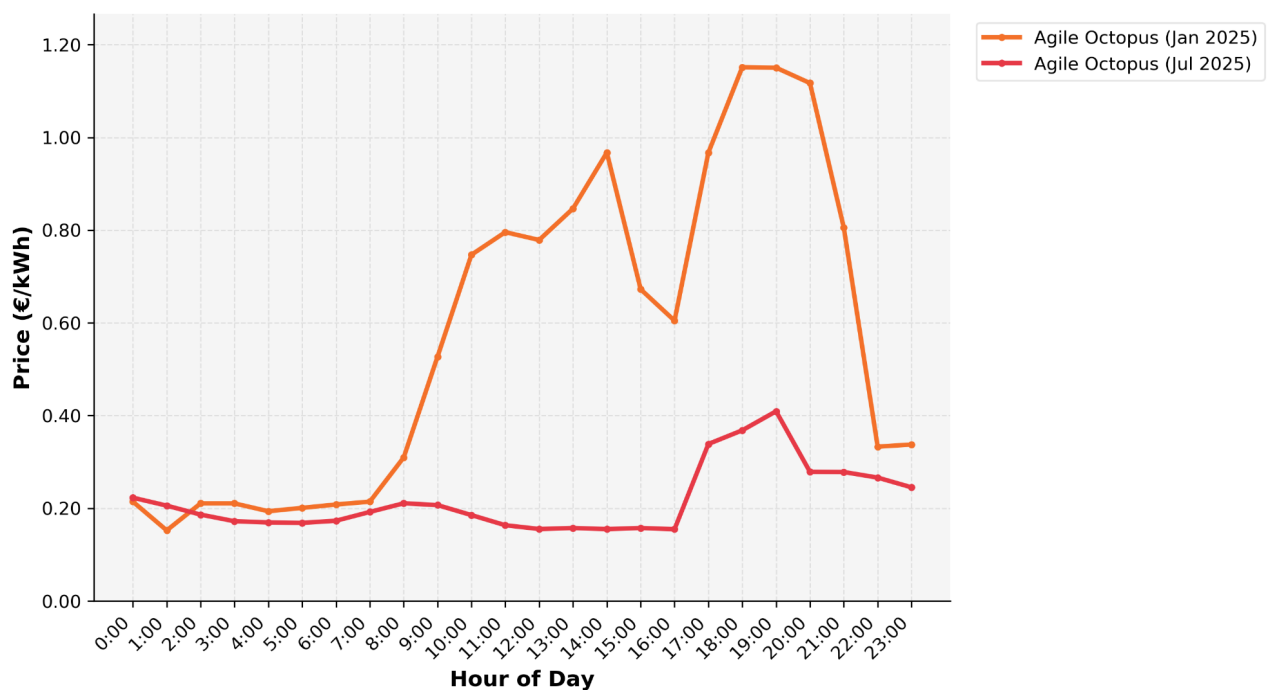


Figure 9. Residential TOU pricing in the UK

¹⁷ Ember (2025)

<https://ember-energy.org/latest-insights/a-record-year-for-british-solar/>

¹⁸ Renewable UK (2025)

<https://www.renewableuk.com/energypulse/blog/stacking-up-the-storage-where-the-uk-battery-market-stands-in-2025/>

2. Overview of Solar plus BESS Financial Model

The solar PV+BESS financial model is a comprehensive tool to calculate the profitability and system-wide impact on the load curve for Vietnam. It draws on real solar insolation data from Vietnam (covering all provinces), and draws on real load curve data from different customers in Vietnam.

2.1. Overview of Input Values and Assumptions

Considerations regarding RTS & BESS dimensioning:

In most cases modelled, there are two conflicting objectives: maximizing the financial return and ensuring that the battery size is large enough to avoid peak electricity tariffs. Since battery costs play a significant role, financial returns typically go down when the battery size goes up. In this analysis, the central analysis has been conducted on the basis of the BESS system enabling the customer to fully avoid peak electricity tariffs (the highest avoided electricity cost). Once this calibration was completed for each customer, a verification was conducted to ensure that the financial returns were still adequate. However, a separate analysis has been included (see Tables 10 - 15) showing the returns of different dimensioning combinations, with small-to-no batteries, and larger solar PV systems. In this way, it is possible to visualize the attractiveness of different dimensioning ratios.

Each customer had to be calibrated to ensure that a suitably-sized solar PV and BESS system is added that matches that specific customer's load curve. The following considerations have been applied to ensure proper dimensioning:

1. For each case (and associated load curve and irradiance curve) **there is a "sweet spot" in terms of RTS dimensioning that the model solves for.** This sweet spot is based on the amount of load that can be directly met with RTS without significant spillage (or curtailment) and it is used to define the total size of the solar array.
2. For the BESS dimensioning, the calibration has been conducted to ensure that the BESS system is large enough to fully avoid the peak pricing in the tariff structure. Since one of the main aims of the analysis is to demonstrate the potential reduction in peak demand from the adoption of BESS, **the battery system is being sized to ensure that both the morning peaks and the evening peaks are largely met either by RTS or BESS** (with sensitivity being provided for a wide range of BESS and RTS sizing). Given the evening peak tariff applies for 3 hours (17:00-20:00), fully avoiding peak prices would lead to a requirement for a 3hr battery. A sensitivity analysis has been conducted to show the returns under different BESS and RTS sizes (see Tables 10 to 15).

A key consideration in the overall analysis is the dimensioning of the solar PV system, as well as the BESS system, in relation to the customer's unique load profile. If the solar PV system is too large, it will lead to high levels of spillage (or curtailment); similarly, an over-dimensioned BESS system is more costly, and may not be optimally used. Also, the duration of the peak prices has an impact on what the "optimal" battery size is: a longer period of peak prices would require a larger battery. Customers therefore face a number of trade-offs in determining the "optimal" size of RTS and BESS to add. The model has been calibrated so that each individual customer has an RTS plus BESS system that matches their individual load curve, and is optimized for the best financial return (IRR) for that case.

Given the adoption of the Direct Power Purchase Agreement (DPPA) framework in Vietnam, which allows customers to purchase power from projects located off-site, **no limitation has been imposed on the maximum solar PV system size in relation to the customer's load, or available rooftop, employee parking lot, or surrounding space.** It is assumed that the solar PV system required can be built to match the load curves provided.

Tariffs:

Regarding the applicable electricity tariffs, the cases modelled are based on the tariffs in place in early 2025 in line with MOIT's Decision No. 1279/QĐ-BCT dated 09 May 2025 on electricity prices. See tariffs in Table 1 above.

Financing Structure:

In terms of financial structure, the modelling was conducted under various scenarios, including a 70:30 Debt:Equity ratio, a 50:50 Debt:Equity ratio, as well as a 0:100 Debt:Equity ratio. However, given that BESS projects are relatively new to banks (not only in Vietnam but also in other markets), it is common for projects that incorporate BESS (and do not rely on a government feed-in tariff, or other off-taker guarantee) to be financed with less debt, and in some cases, without any debt at all. Many commercial customers prefer to rely on balance sheet financing rather than take on debt to finance BESS projects in particular. One of the primary reason is, as highlighted further below, BESS projects in Vietnam today (under the single-part/energy-only electricity tariff) do not generate positive financial returns on a stand-alone basis: for a bank to be requested to provide debt financing, project developers would need to be able to demonstrate some additional benefit to the customer from adopting BESS, such as improved electricity reliability, or the generation of additional revenues (such as from ancillary services such as frequency control).

Since electricity supply remains highly reliable in Vietnam, and BESS projects cannot yet earn revenues from ancillary services provision, these options do not enter the picture from a lender's perspective. As such, in order to lend to a customer specifically for adding BESS, a bank would be offering the credit on the basis of the customer's creditworthiness, not the attractiveness of the BESS project as such. In fact, if loss-making, a BESS project could even make a borrower less creditworthy from a bank's perspective.

The project results present project-specific IRRs. By using IRRs, a more accurate picture of the overall financial attractiveness of the investments is established.

Financing Assumptions	Values
Debt Share	0% / 50% / 70%
Self-financed Share (Equity)	100% / 50% / 30%
Cost of Debt	6.43%
General Inflation (CPI)	4.75%
Tax Rate	20%
Operating Life (Solar PV)	25 years
CAPEX for Solar PV	USD \$400/kW (VND 10.5 million)
OPEX for Solar PV	USD \$15/kW per year ¹⁹
Degradation for Solar PV	0.4% per year
CAPEX for BESS	USD \$220/kWh
Operating Life (BESS)	15 years
Cost of Repowering BESS (after year 15)	USD \$100/kWh
OPEX for BESS	USD \$10/kWh per year ²⁰
Degradation for BESS	2% per year, starting year 2

Table 2. Summary of Key Financing and Modelling Assumptions

Initially, the modelling was conducted with a 70:30 Debt:Equity ratio as well as on a 50:50 basis, as these were the common financing structures used under the previous Feed-in Tariff projects. However, since the FIT has been discontinued and most commercial and industrial solar PV RTS investments are mainly financed on-balance sheet, additional sensitivities have been conducted to examine cases without debt financing (i.e., 0:100 Debt:Equity ratio).

¹⁹ NREL ATB analysis: https://atb.nrel.gov/electricity/2024/commercial_pv

²⁰ Lazard June 2025: <https://www.lazard.com/media/uounhon4/lazards-lcoeplus-june-2025.pdf>

2.2. Overview of Load Curves

The chart below shows the typical total power system demand curve in Vietnam for 2023, compared to the current TOU tariff structure.

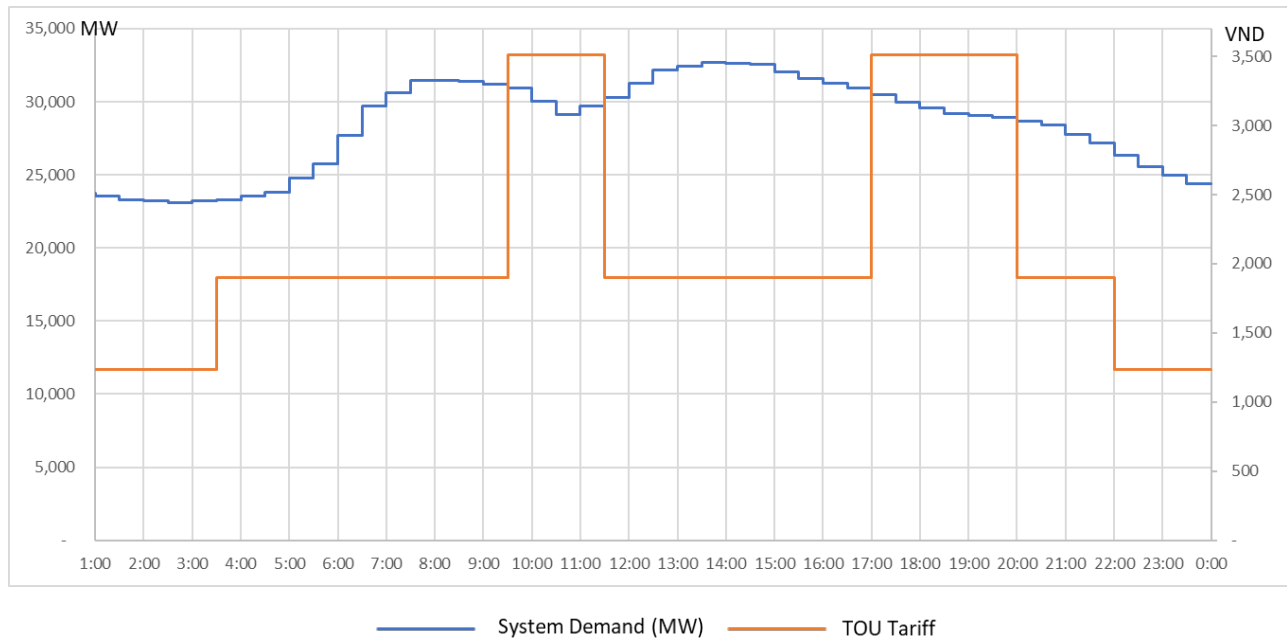


Figure 10. Viet Nam's Typical Load Profile (30' interval) vs TOU tariffs

As can be seen above (Figure 10), the overall system load curve indicates a slight decrease in demand between 9:30AM and 11:30AM that corresponds to the TOU rates in place; a similar, but less prominent, decline in demand is seen in the evening hours (between 17:00 and 20:00). In many other countries, the total system peak occurs in the evening hours, and is driven by household electricity consumption from cooking, air conditioning, and other household demand. In Vietnam, by contrast the load curve is more heavily shaped by commercial and industrial demand, with the household sector playing a more modest role.

As can be seen above, the TOU tariff appears to have already functioned in shifting the load curve away from the evening peaks, which suggests that further adjustments to TOU rates can play a meaningful role in determining the aggregate system load shape. The chart below (Figure 11) shows that the commercial and industrial sectors in particular have already adjusted their demand significantly in line with TOU rates, which shows an even greater responsiveness to TOU tariffs.

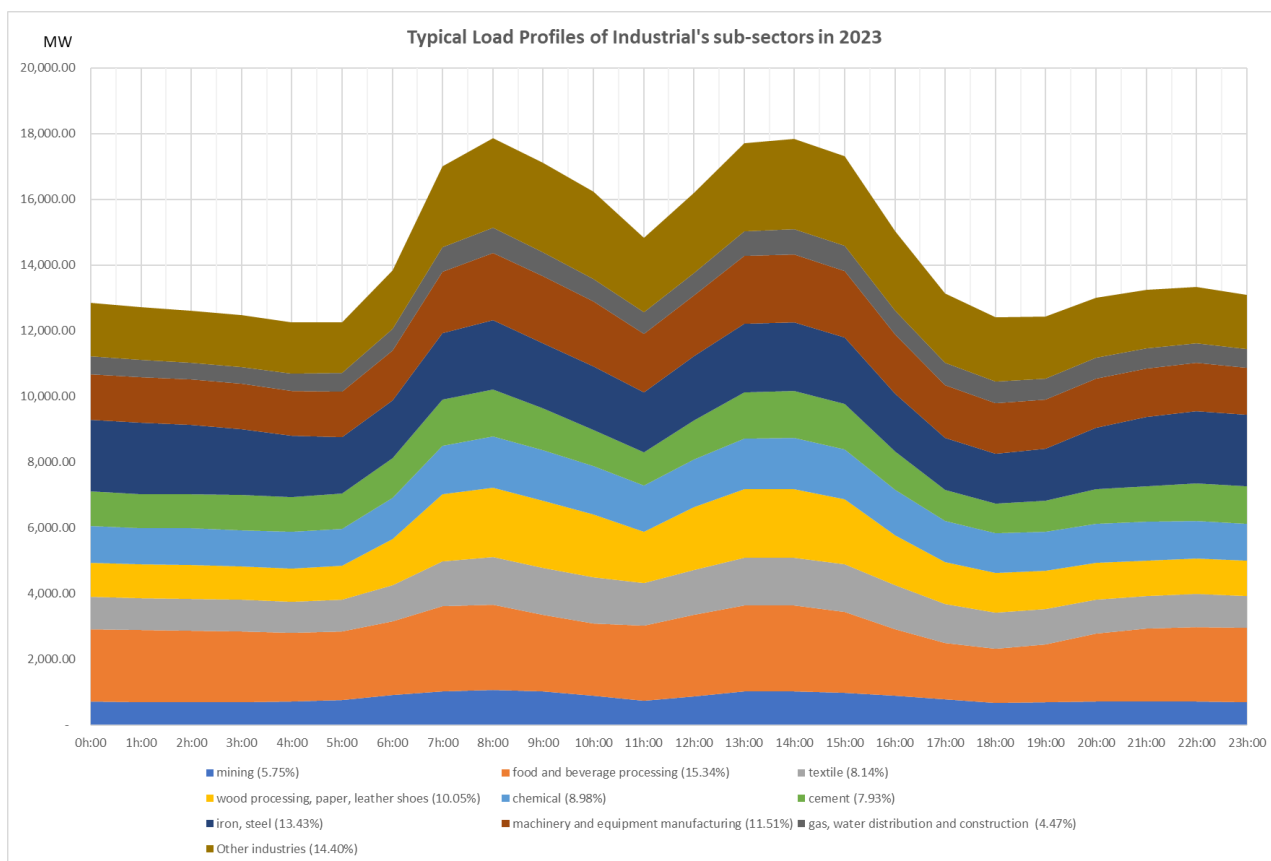


Figure 11. Typical Load Profiles of Industrial's sub-sectors in 2023

Data Source: MOIT & EVN 2024

No.	Sub-sectors	Share (%)	Targeted groups
1	Mining	5.7%	
2	Food and beverage processing	15.3%	x
3	Textile	8.1%	(same profile as 8)
4	Wood processing, paper, leather shoes	10.0%	(same profile as 8)
5	Chemical	9.0%	(same profile as 8)
6	Cement	7.9%	
7	Iron, steel	13.4%	x
8	Machinery and equipment manufacturing	11.5%	x
9	Gas, water distribution and construction	4.5%	
10	Other industries	14.4%	

Table 3. Share of Sub-sectors to the Typical Load profile of Industrial Sector

Data Source: MOIT & EVN 2024

As seen above, the main customer categories chosen were selected based on the fact that they represent the customer categories with the largest individual share of total commercial and industrial power demand in Vietnam. Also, many of the other associated sectors (i.e., textiles, wood processing, etc.,) have load profiles that are quite similar to those of the customer categories selected. As such, the analysis aims to reflect the “real” potential impact of solar plus BESS adoption among commercial and industrial customers in Vietnam.

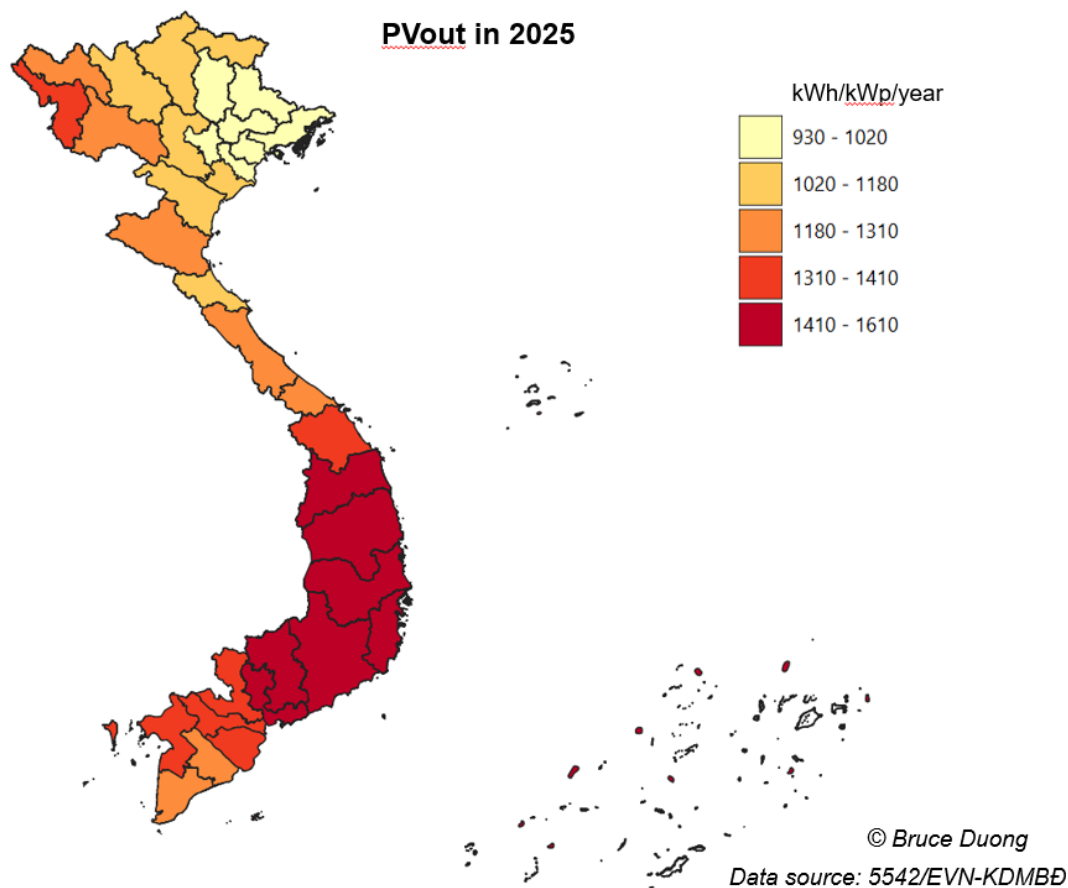


Figure 12. Map of annual PV output attribution in Viet Nam

Graphic source: GLZ built based on input from 5542/EVN-KDMBD

The map above shows Vietnam’s solar resource quality by region. To date, much of the solar power in Vietnam has been concentrated in the sunnier southern regions of the country. However, in order to advance the energy transition in Vietnam, and increase power supply in the northern parts of the country, the Government is keen to encourage more solar development in the North. The development of RTS on commercial and industrial rooftops can help contribute to this strategic objective, while the addition of storage can have further beneficial impacts on the load curve in the north while improving power reliability and energy security for customers.

With regard to individual customer loads, the model that has been developed uses real load curves from individual customers in Vietnam provided by EVN. **The load curves are anonymized and cannot be traced to a particular customer or industrial site.** They are meant to reflect a representative customer in that commercial or industrial customer category.

Note on the selection of the load curves: there is no such thing as a perfect load curve, or a perfectly “representative” load curve. Each load curve for each individual customer has unique characteristics, peaks and valleys that change by day, by week, by month, and by season. The selection of the load curves was done based on the load curves available, and benchmarked against the sector-wide load curves for those sector types. The analysis was run on the basis of these real customer load curves.

Sectors and Cases Analyzed:

Four sectors were identified (see *above justification*): steel/metallurgy, food processing, machinery, and hotels. The analysis was conducted on each sub-sector to model this customer type, and its adoption of RTS plus BESS was based on a case-by-case dimensioning of a suitable solar PV and BESS system for that specific customer, including that specific load curve (see above).

In order to arrive at an “average” load curve impact for a 10%, 25%, or 50% RTS plus BESS penetration rate for each customer type, the load curve size was re-dimensioned to represent an “average” customer of that type. This is explained in further detail below.

The results below show the impacts of different levels of adoption of RTS plus BESS by different customer groups in Vietnam, using locations in both Northern Vietnam and Southern Vietnam (specifically, Bac Ninh in Northern Vietnam, and Ho Chi Minh City, District 1, for Southern Vietnam). **Note:** The impacts are modelled on the total system demand curve for Vietnam (not just the North, or the South).

Methodology to arrive at an estimate of the system-level impacts on the load curve:

For each customer, the individual load curve was summed over the year to arrive at a total annual electricity demand for that customer. The customer’s total electricity demand was then compared to the average power demand per customer for that industrial sub-sector, and a multiplier was applied to arrive at a load impact for that customer that represents an “average” customer load of the average customer for that sub-sector (i.e., food processing, steel, machinery, and hotel). This “average” customer was then used to model the impact of 10%, 25% and 50% of all customers of that type adopting RTS plus BESS on the whole Vietnam power system.

For the cases highlighting the impacts of a 50% penetration rate, this is considered to represent a “high” case of adoption of RTS plus BESS in Vietnam. It is important to note that this is **not a forecast**: it is an output that can be used to examine the impacts of different levels of RTS plus BESS adoption on the overall system load curve. By examining a 50% adoption scenario, it is possible to envision what the total impact on the VN power demand curve would be under a high RTS plus BESS adoption future.

More detailed analysis could be conducted to examine the impact only on specific regions of Vietnam, for instance, a 30% penetration rate among all hotels in northern Vietnam, applied against the NPC load curve. For the time being, this analysis concentrates on the Vietnam-wide impacts of RTS plus BESS adoption for the entire load curve of Vietnam. Further analysis (e.g., just on NPC, or on EVN-HCMC, would need to be conducted separately).

3. Analysis of Impact on System Load

The analysis focuses on four main customer types, including three of the largest individual sub-categories of industrial power demand in Vietnam, plus hotels. The following sub-sections cover the following customer types in the following order:

- ❖ **Steel and metallurgy**
- ❖ **Food processing**
- ❖ **Hotels**
- ❖ **Machinery-related manufacturing**

Each sector's load curve is based on a representative load curve that broadly matches the sector-wide load curve for that customer type in Vietnam. The following table shows the annual demand of each customer type modelled:

Customer Type	Annual Load	Monthly Demand (in MWh)																										
Steel and metallurgy	244,655 MWh per year (244.7 GWh)	<table><tr><th>Month</th><th>Monthly Demand (MWh)</th></tr><tr><td>1</td><td>16,924</td></tr><tr><td>2</td><td>11,126</td></tr><tr><td>3</td><td>16,504</td></tr><tr><td>4</td><td>18,785</td></tr><tr><td>5</td><td>17,525</td></tr><tr><td>6</td><td>21,128</td></tr><tr><td>7</td><td>28,205</td></tr><tr><td>8</td><td>29,646</td></tr><tr><td>9</td><td>25,279</td></tr><tr><td>10</td><td>21,904</td></tr><tr><td>11</td><td>19,387</td></tr><tr><td>12</td><td>18,242</td></tr></table> Monthly load profile (mWh)	Month	Monthly Demand (MWh)	1	16,924	2	11,126	3	16,504	4	18,785	5	17,525	6	21,128	7	28,205	8	29,646	9	25,279	10	21,904	11	19,387	12	18,242
Month	Monthly Demand (MWh)																											
1	16,924																											
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3	16,504																											
4	18,785																											
5	17,525																											
6	21,128																											
7	28,205																											
8	29,646																											
9	25,279																											
10	21,904																											
11	19,387																											
12	18,242																											
Food processing	1,966 MWh per year (2.0GWh)	<table><tr><th>Month</th><th>Monthly Demand (MWh)</th></tr><tr><td>1</td><td>141</td></tr><tr><td>2</td><td>116</td></tr><tr><td>3</td><td>179</td></tr><tr><td>4</td><td>173</td></tr><tr><td>5</td><td>186</td></tr><tr><td>6</td><td>179</td></tr><tr><td>7</td><td>152</td></tr><tr><td>8</td><td>179</td></tr><tr><td>9</td><td>178</td></tr><tr><td>10</td><td>164</td></tr><tr><td>11</td><td>165</td></tr><tr><td>12</td><td>155</td></tr></table> Monthly load profile (mWh)	Month	Monthly Demand (MWh)	1	141	2	116	3	179	4	173	5	186	6	179	7	152	8	179	9	178	10	164	11	165	12	155
Month	Monthly Demand (MWh)																											
1	141																											
2	116																											
3	179																											
4	173																											
5	186																											
6	179																											
7	152																											
8	179																											
9	178																											
10	164																											
11	165																											
12	155																											
Hotels	1,313 MWh per year (1.3GWh)	<table><tr><th>Month</th><th>Monthly Demand (MWh)</th></tr><tr><td>1</td><td>74</td></tr><tr><td>2</td><td>79</td></tr><tr><td>3</td><td>88</td></tr><tr><td>4</td><td>123</td></tr><tr><td>5</td><td>113</td></tr><tr><td>6</td><td>152</td></tr><tr><td>7</td><td>149</td></tr><tr><td>8</td><td>149</td></tr><tr><td>9</td><td>106</td></tr><tr><td>10</td><td>93</td></tr><tr><td>11</td><td>98</td></tr><tr><td>12</td><td>90</td></tr></table> Monthly load profile (mWh)	Month	Monthly Demand (MWh)	1	74	2	79	3	88	4	123	5	113	6	152	7	149	8	149	9	106	10	93	11	98	12	90
Month	Monthly Demand (MWh)																											
1	74																											
2	79																											
3	88																											
4	123																											
5	113																											
6	152																											
7	149																											
8	149																											
9	106																											
10	93																											
11	98																											
12	90																											

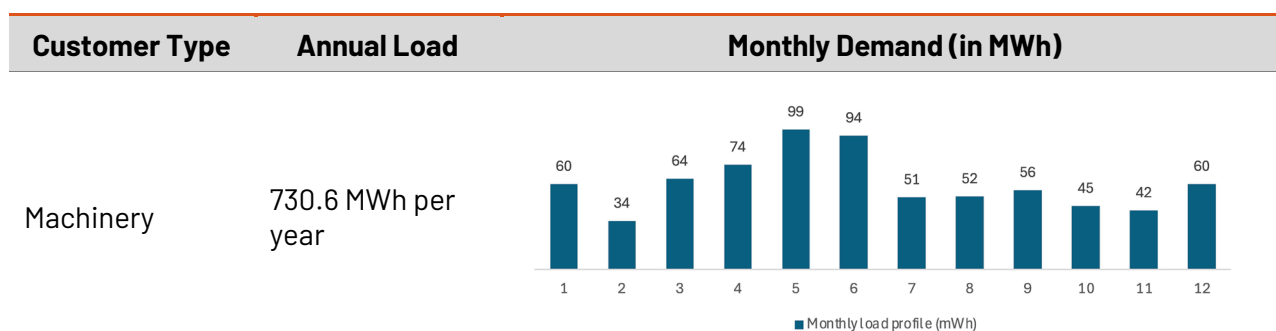


Table 4. Summary of Annual and Monthly Demand by Customer Type

Data Source: MOIT & EVN 2024

After analyzing the impacts on the overall system load of the power system, the following section provides an overview of the financial attractiveness of the various RTS and BESS cases (Section 4) before turning to the conclusions and recommendations (Section 5).

Sub-sector	Food processing	Machinery & equipment manufacturing	Steel	Hotel & Hospitation
Name in Annual Statistical Book 2024	Manufacture of Food product & Manufacture of beverage	Manufacture of computers, electronic and optical products & Manufacture of electrical equipment & Manufacture of machinery and equipment & Motor vehicles, trailers and semi-trailers & Other transport equipment & Others	Manufacture of basic metal	Accommodation
# of Consumers	12,087	11,465	1,635	10,673

Table 5. Number of Target customers

Source: Annual statistical book 2024 (GSO – MPI)²¹

²¹ https://www.nso.gov.vn/wp-content/uploads/2025/06/NGTK-Cuc-TK-2024_BQ.pdf

3.1. Steel and Metallurgy Sector

Total national power demand in Vietnam stood at 251TWh in 2022 (which has increased to 276.4TWh in 2024). Based on 2022 figures, which are used to show the customer shares below, according to government figures, industrial sector demand represents 51.1% of national demand. This means that **the industrial sector as a whole** (which includes a wide-range of sub-sectors including machinery, food processing, metallurgy, accommodation, and others) **represents roughly 127 TWh of annual power demand**.

According to the Annual Statistical Book (2024) which shows data for Vietnam's power sector in 2022, the metallurgy sector in Vietnam comprises **1.635 customers** in total representing **13.43% of total industrial power demand**. At 13.43%, this means that the steel/metallurgy sector consumes approximately 17 TWh of electricity per year in Vietnam.

With 1.635 customers in total in this category, the "average" metallurgy customer therefore consumes roughly 10GWh per year.

The chart below shows the monthly load profile used for an individual customer in this subsector; it is from a steel and metallurgy customer in northern Vietnam.

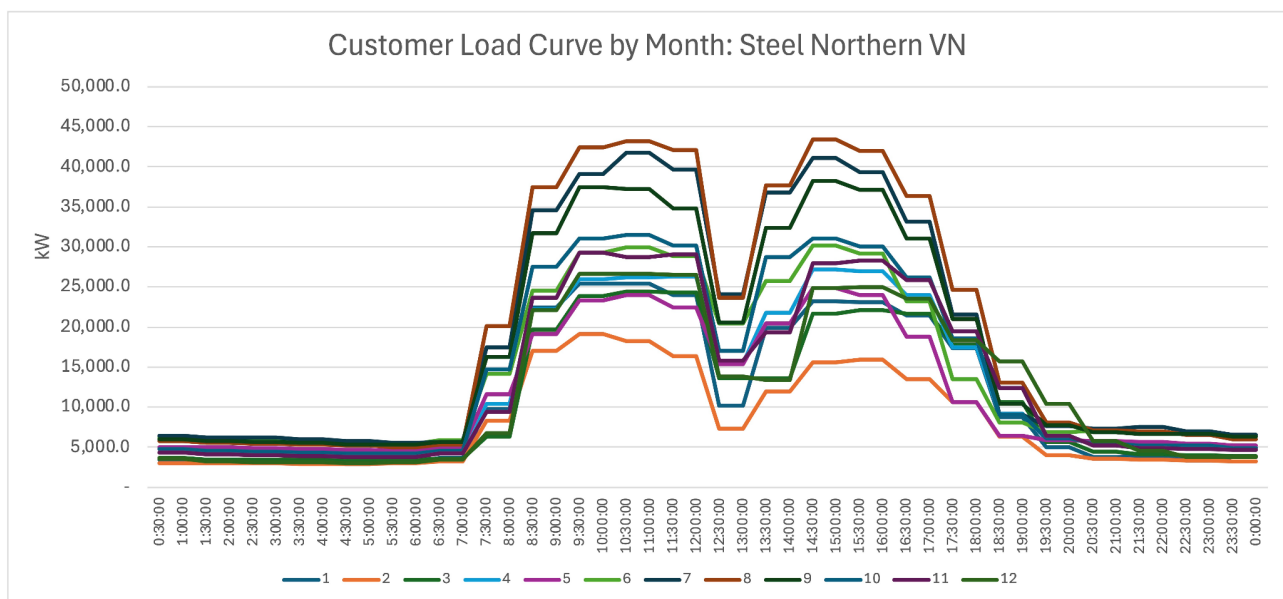


Figure 13. Monthly Steel/Metallurgy Load Curve from Northern Vietnam

In each customer case that adds solar PV+BESS, the solar PV system has been dimensioned to optimize the use of solar, and to avoid the TOU rates. More specifically, the timing of the battery charging-and-discharging is driven by the TOU tariffs in place for that customer type and voltage level.

The chart shows the daily profile of solar PV and BESS use for a **steel and metallurgy customer** located in northern Vietnam:

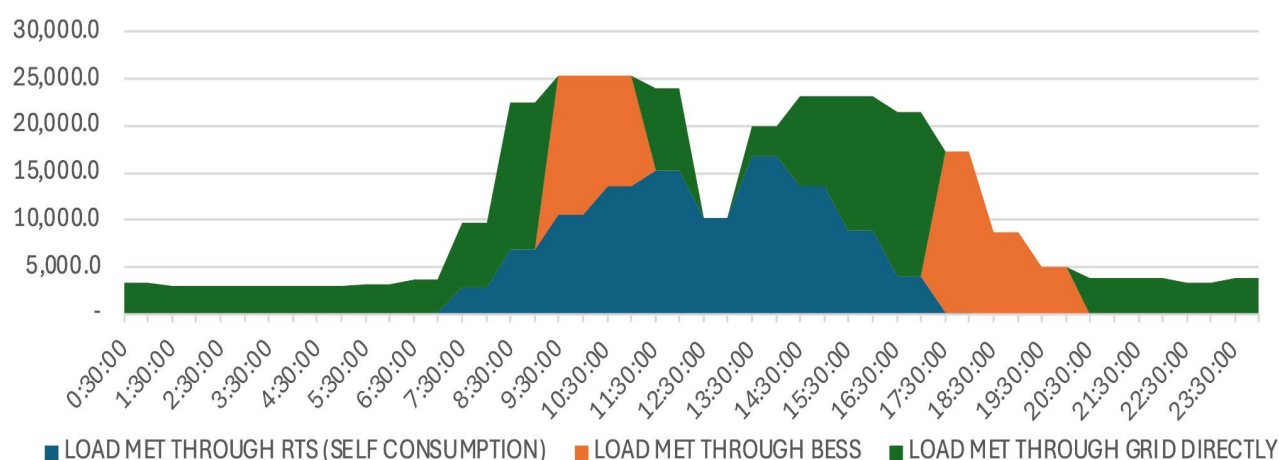


Figure 14. Daily profile of RTS plus BESS on a typical day: steel/metallurgy customer

As can be seen, the RTS system for this user is considerably smaller than it might otherwise be, considering the customer's overall electricity demand: the reason is mainly due to the shape of the load curve, specifically the sharp drop in demand around midday. For this customer type and load curve, having a large or "over-dimensioned" solar PV system would lead to significantly higher spillage, or curtailment. As a result, the PV system has been dimensioned more conservatively to avoid excessive electricity spillage.

As can also be seen in the case of the steel producer, the BESS system is used to meet demand both in the morning hours and in the evening hours, as this is economically advantageous (for this user type, avoiding the on-peak tariffs generates significant savings due to its high morning demand). Given this unique load profile and RTS+BESS usage pattern, and the fact that individual steel/metallurgy customers have a large load, the adoption of RTS+BESS among these customers has a significant impact on the overall system load curve.

The charts below show the impact of solar PV+BESS adoption in the metallurgy sector under different penetration levels (10%, 25%, and 50%).

10% RTS plus BESS Penetration Rate:

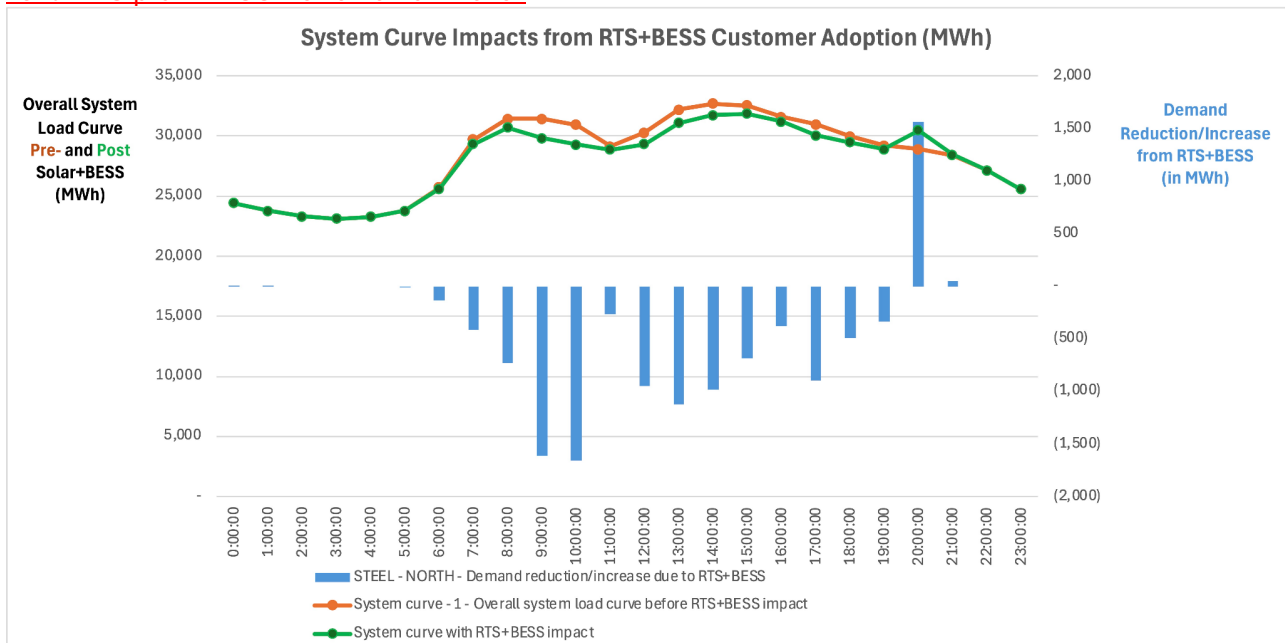


Figure 15. System-level impact of Steel/Metallurgy

25% RTS plus BESS Penetration Rate:

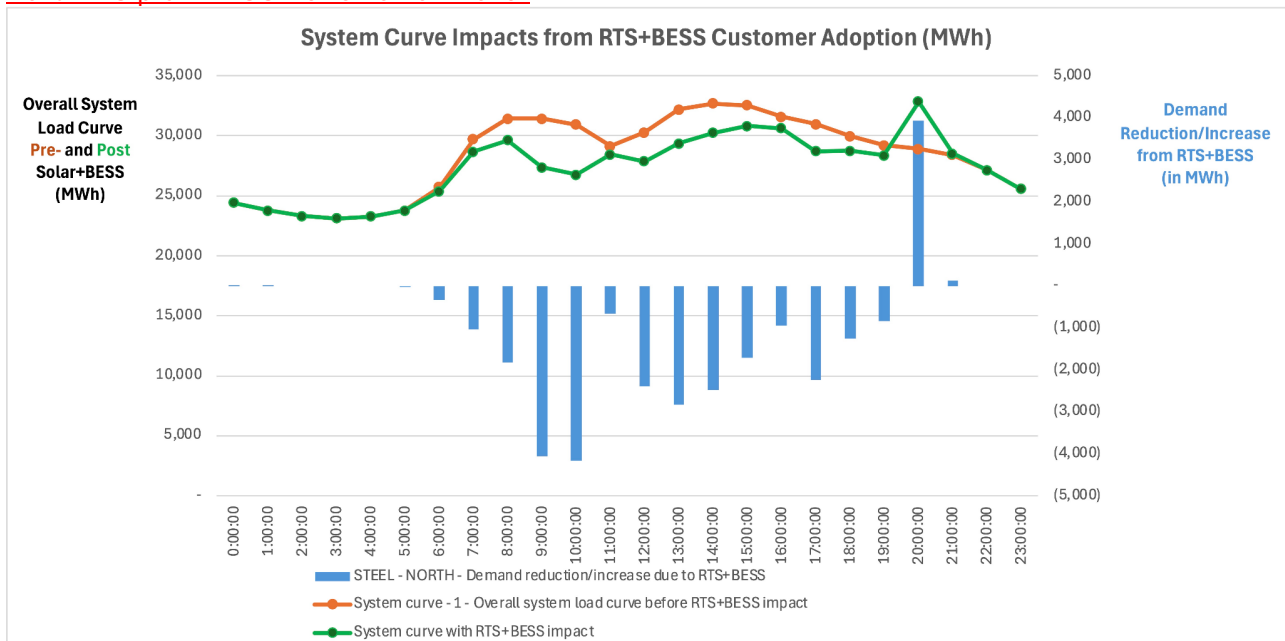


Figure 16. System-level impact of Steel/Metallurgy

50% RTS plus BESS Penetration Rate:

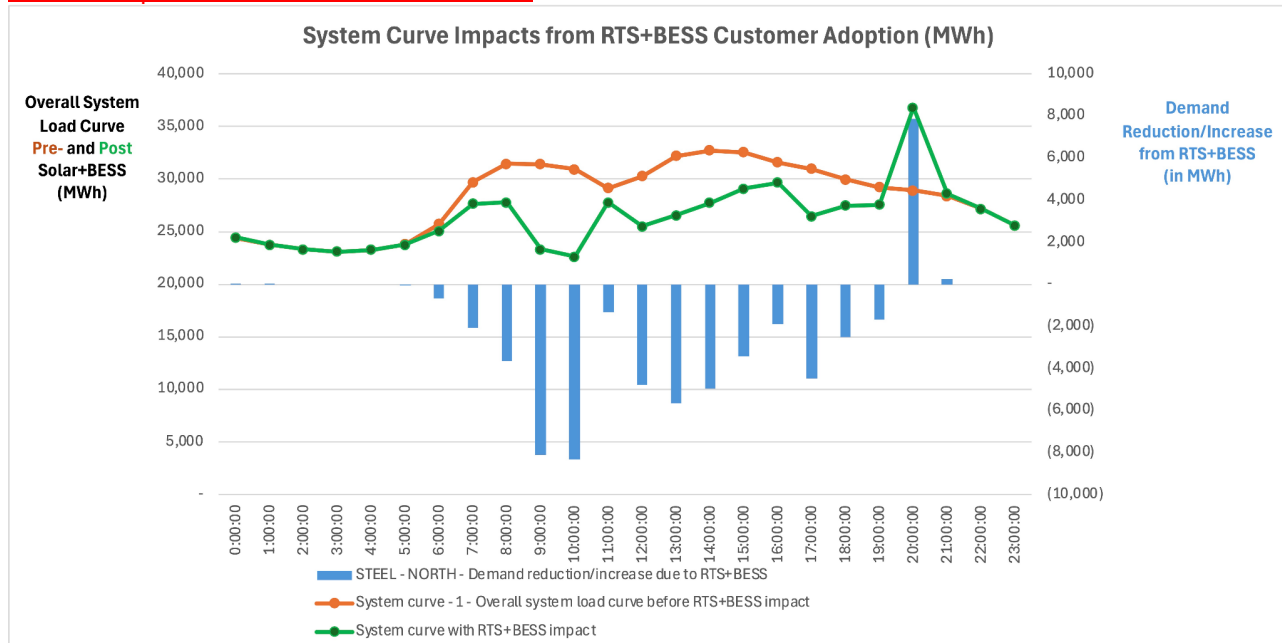


Figure 17. System-level impact of Steel/Metallurgy

Key Findings from Steel/Metallurgy Sector

- Despite the fact that the total number of customers in this category is relatively small at 1.635 in total, each customer has a fairly large demand. As a result, the impacts of a few hundred customers adopting solar PV+BESS can be quite large.
- **Indeed, the overall impact on the power system demand curve is quite significant:** at a 50% RTS+BESS penetration rate, the reduction on system load reaches **8,300MW** during the daytime.
- In terms of its contribution to mitigating peak electricity demand, the adoption of RTS+BESS in the metallurgy sector among roughly 50% of the sector's customers would trigger a **15% - 25% reduction** in peak electricity demand in Vietnam.
- Due to their tariff category and the TOU pricing in place, the model shows that it would be economically advantageous for metallurgy customers to draw on solar and onsite storage during the morning and afternoon hours, but to benefit from the mid-day hours at the "normal" tariff level to re-charge the battery in between. This can be seen in the midday spike in demand shown above, and in the different impact on the overall load curve compared to other customer types.
- However, based on Vietnam's current TOU tariff, which declines in the evening, **a clear demand spike can be seen in the evening hours.** At this point, BESS systems are activated and start charging during the "off-peak" tariff hours. Given the current structure and timing of Vietnam's TOU tariffs, a spike in power demand can therefore be expected at this time. This applies to all RTS+BESS customers covered in this report. This spike could be mitigated or delayed via regulation, via mandate, or via changes to the TOU tariff structure.

3.2. Food Processing Sector

According to the Annual Statistical Book (2024) which shows data for Vietnam's power sector in 2022, the food processing sector in Vietnam comprises **12.087 customers** in total representing **15.35% of industrial electricity demand**. This makes food processing one of the large commercial and industrial sectors in Vietnam in terms of power demand. Given total industrial sector demand of 127TWh, this means that the food processing and beverage sector represents approximately 19.49TWh of annual power demand in Vietnam.

The chart below shows the monthly load profile used for the food processing sector. It comes from a food processing customer in southern Vietnam.

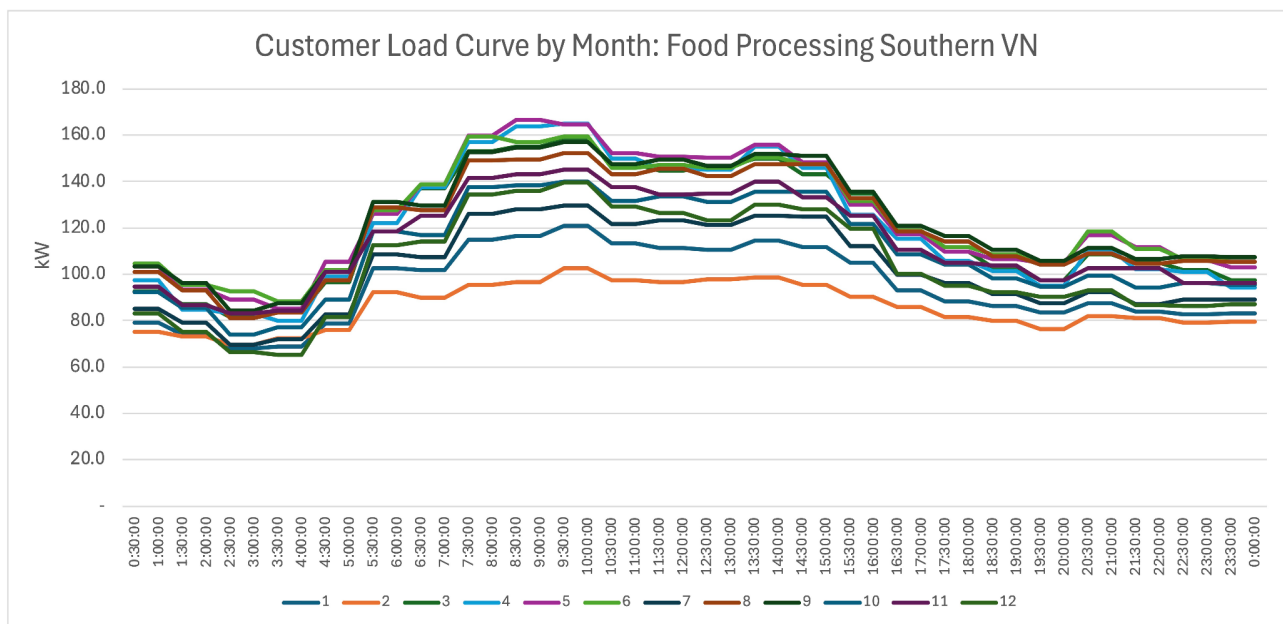


Figure 18 Monthly Food Processing Load Curve from Southern Vietnam

In each customer case that adds solar PV+BESS, the solar PV system has been dimensioned to optimize the use of solar, and the timing of the battery charging-and-discharging is driven by the TOU tariffs in place for that customer type and voltage level.

The chart shows the daily profile, including the solar and BESS usage, for a **food processing customer** located in southern Vietnam:

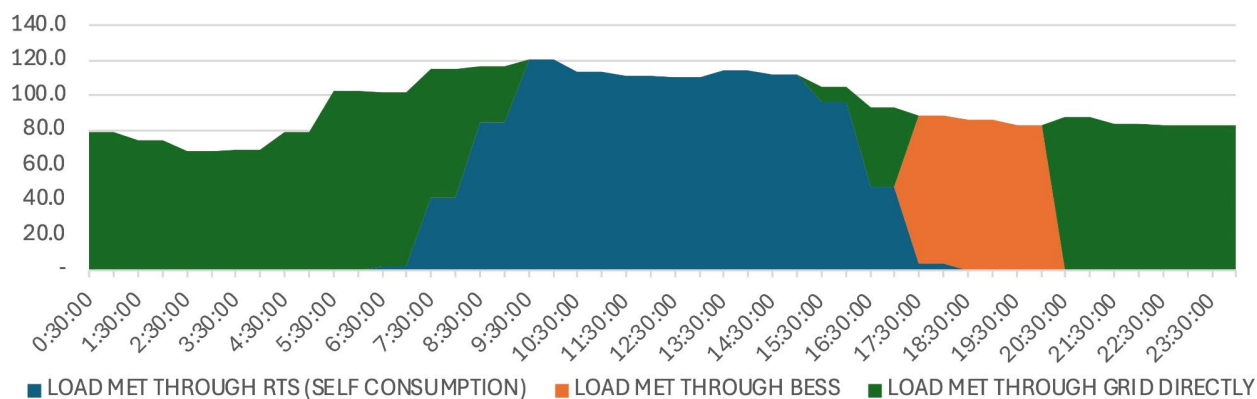


Figure 19 Daily profile of RTS+BESS on a typical day: Food processing sector

As can be seen, in the case of the food processing sector, the solar PV system is used to meet daytime demand and charge the battery, with no activation of the battery to discharge during the daytime. This is due to the shape of the load curve that enables a better matching of PV output to real-time demand: with a well-dimensioned solar PV system, a customer with this load curve could regularly avoid peak pricing with its solar system alone. On cloudier or rainier days, the BESS system (recharged during the overnight off-peak period) would be activated. In the case of the food processing load curve, the BESS system shown here only discharges in the evening hours.

10% RTS+BESS Penetration Rate:

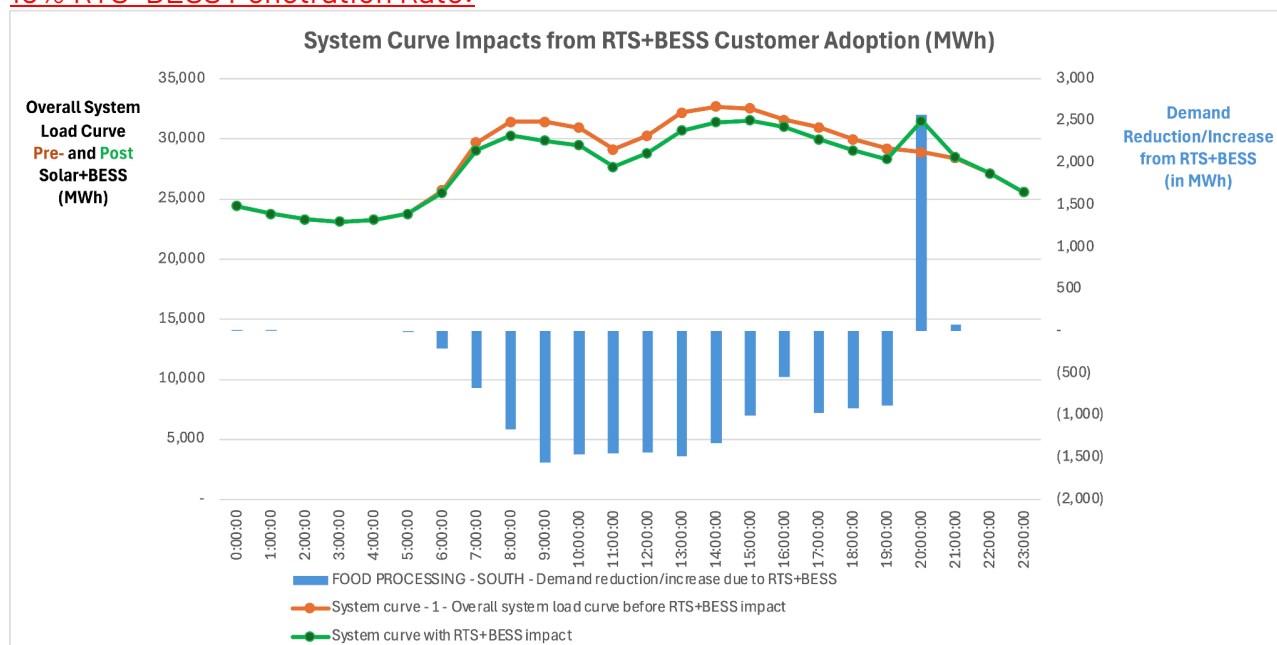


Figure 20. System-level impact of Food Processing

25% RTS+BESS Penetration Rate

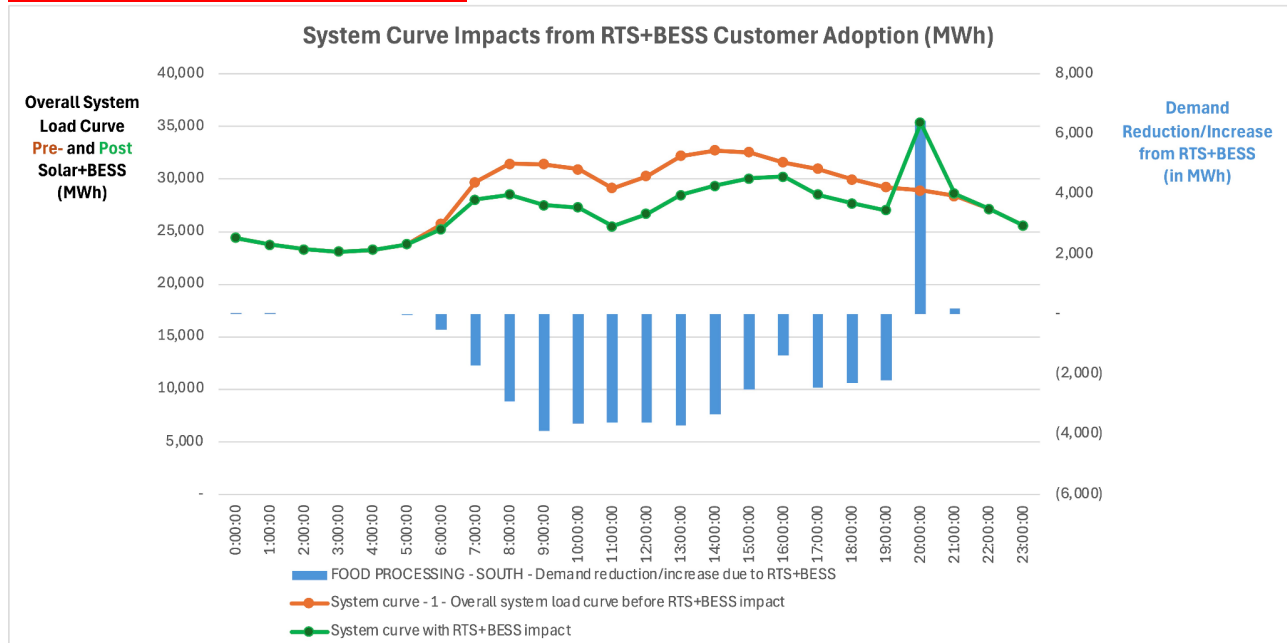


Figure 21. System-level impact of Food Processing

50% RTS+BESS Penetration Rate:

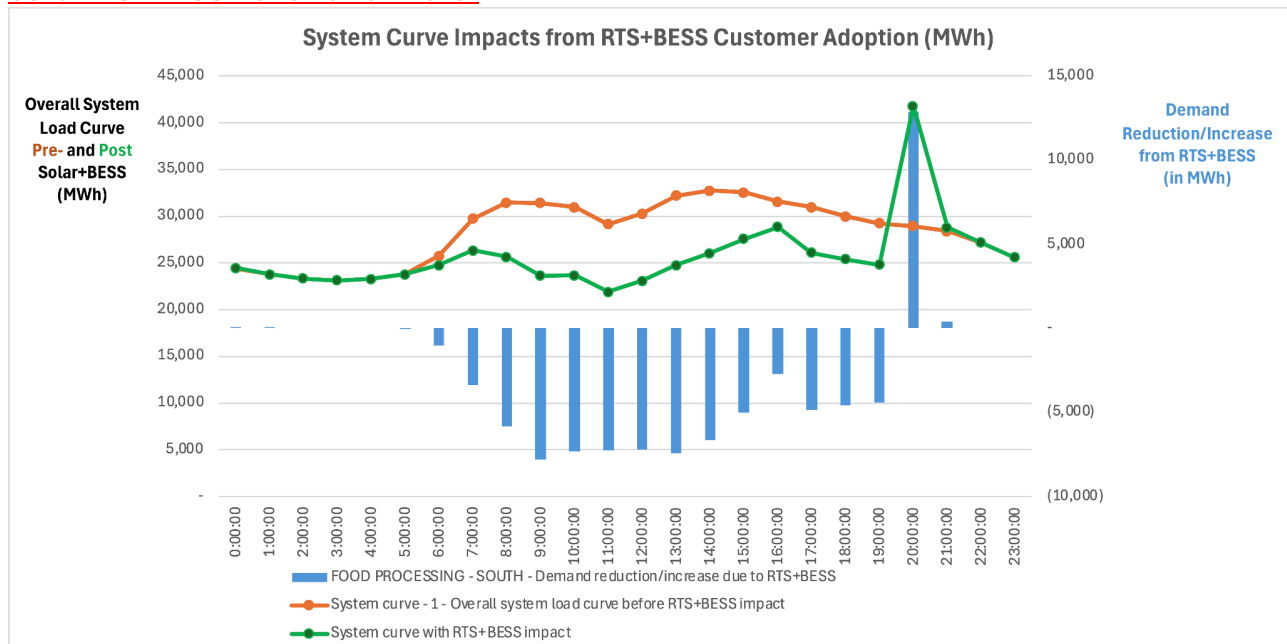


Figure 22. System-level impact of Food Processing

Key Findings from RTS+BESS adoption in the **Food Processing** sector

- **The overall impact on the power system demand curve is significant, particularly at higher levels of penetration:** at an estimated 50% RTS+BESS penetration, the aggregate impact on system load is around **7,500MW** during the daytime.
- In terms of its contribution to mitigating peak electricity demand, the adoption of RTS+BESS in the food processing sector among roughly 50% of the sector's customers would trigger a **15% - 25% reduction** in peak electricity demand.
- However, based on Vietnam's current TOU tariff, which declines in the evening, a **clear demand spike can be seen in the evening hours**. At this point, BESS systems are activated and start charging during the "off-peak" tariff hours. Given the current structure and timing of Vietnam's TOU tariffs, this could lead to a spike in power demand at this time. This spike could be mitigated or delayed via regulation, via mandate, or via changes to the TOU tariff structure.

3.3. Hotel Sector

This section examines hotel customers using a customer profile from southern Vietnam.

According to the Annual Statistical Book (2024) which shows data for Vietnam's power sector in 2022, the hotel sector in Vietnam comprises **10.673 customers**. However, in contrast to the Food Processing industry, which represents over 13% of industrial power demand, the hotel industry only represents **1.52% of industrial demand** (or 1.9TWh of annual power demand). This makes the hotel industry in Vietnam a relatively small contributor in terms of overall electricity demand.

The chart below shows the hotel demand curve by month used for the modelling:

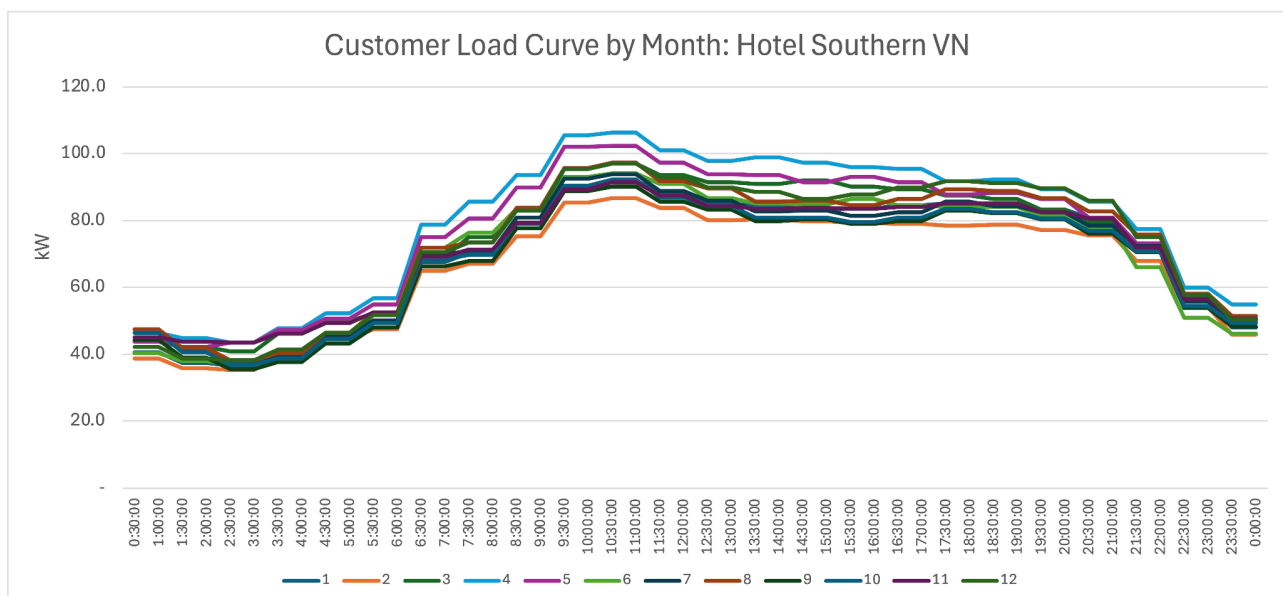


Figure 23 Monthly Hotel Load Curve from Southern Vietnam

In each customer case that adds solar PV+BESS, the solar PV system has been dimensioned to optimize the use of solar, and the timing of the battery charging-and-discharging is driven by the TOU tariffs in place for that customer type and voltage level. The chart shows the daily profile for a **hotel customer** located in southern Vietnam:

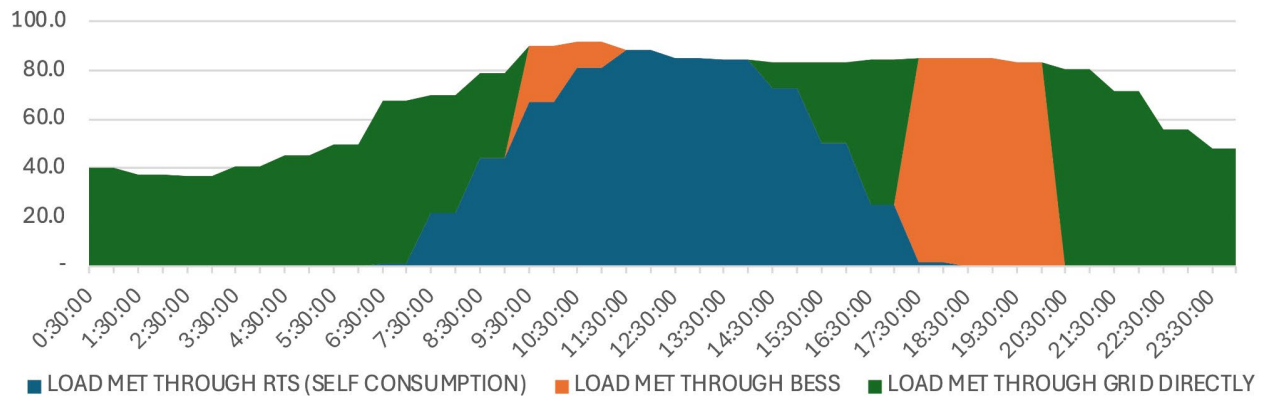


Figure 24. Daily profile of RTS+BESS on a typical day of Hotel customer

Due to the hotel customer's load curve, which starts to rise already around 5:30AM - 6:00AM, the BESS system starts to discharge as soon as peak pricing starts. By doing so, it helps the customer avoid having to pay the peak tariffs.

10% RTS+BESS Penetration Rate:

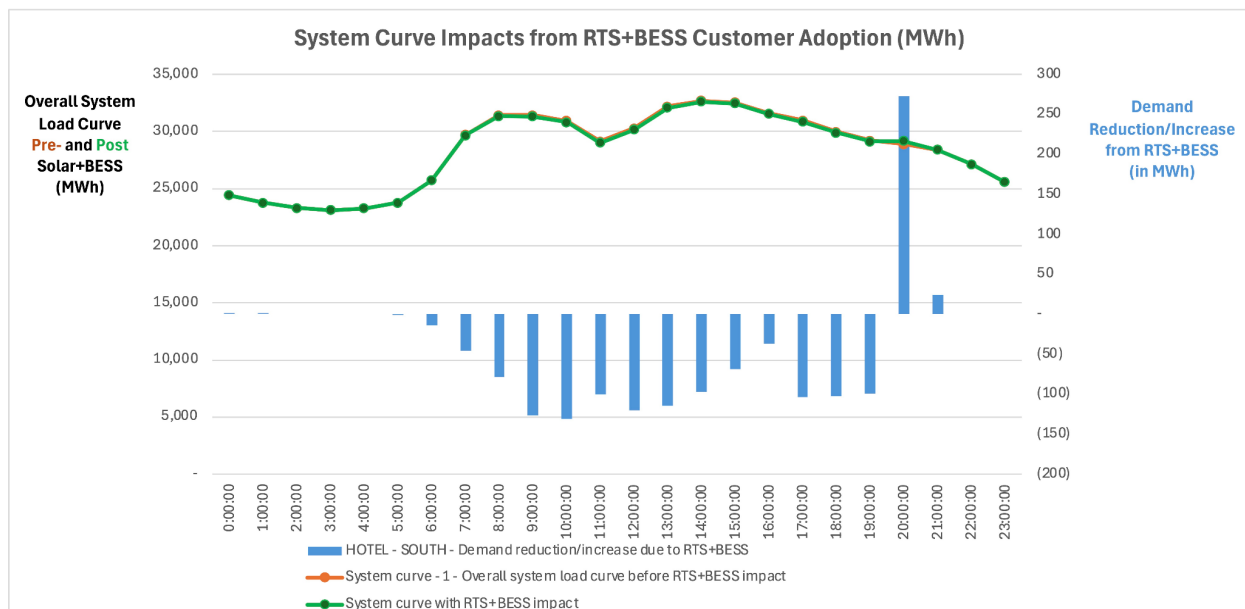


Figure 25. System-level impact of Hotel customers

(Note: Given the small size of the hotel impact, the 25% penetration scenario has been excluded)

50% RTS+BESS Penetration Rate:

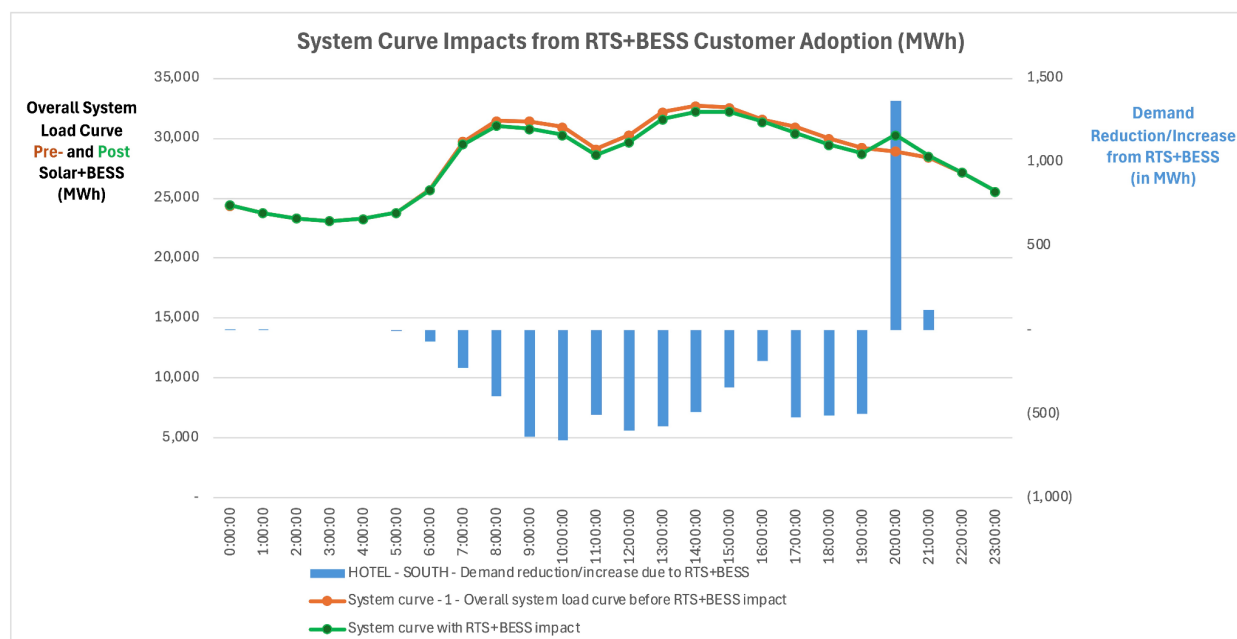


Figure 26. System-level impact of Hotel customers

Key Findings from RTS+BESS adoption in the **Hotel** sector

- **The overall impact on the power system demand curve is relatively small, even at higher levels of penetration:** at an estimated 50% RTS+BESS penetration, the aggregate impact on system load is only around **650MW** during the daytime.
- In terms of its contribution to mitigating peak electricity demand, the adoption of RTS+BESS in the hotel sector among roughly 50% of the sector's customers would trigger a **1.5% - 2% reduction** in peak electricity demand.

However, based on Vietnam's current TOU tariff, which declines in the evening, a **clear demand spike can be seen in the evening hours**. At this point, BESS systems are activated and start charging during the "off-peak" tariff hours. Given the current structure and timing of Vietnam's TOU tariffs, this could lead to a spike in power demand at this time. This spike could be mitigated or delayed via regulation, via mandate, or via changes to the TOU tariff structure.

3.4. Machinery Sector

This section examines the machinery sector using a customer profile from northern Vietnam.

According to the Annual Statistical Book (2024) which shows data for Vietnam's power sector in 2022, the machinery sector in Vietnam comprises **11.465 customers representing 11.51% of total industrial electricity demand (or 14.6 TWh of annual demand)**. This makes the machinery sector in Vietnam an important contributor in terms of overall power demand.

The chart below shows the monthly load curves for the machinery and equipment manufacturer customer selected. The customer is based in northern Vietnam.

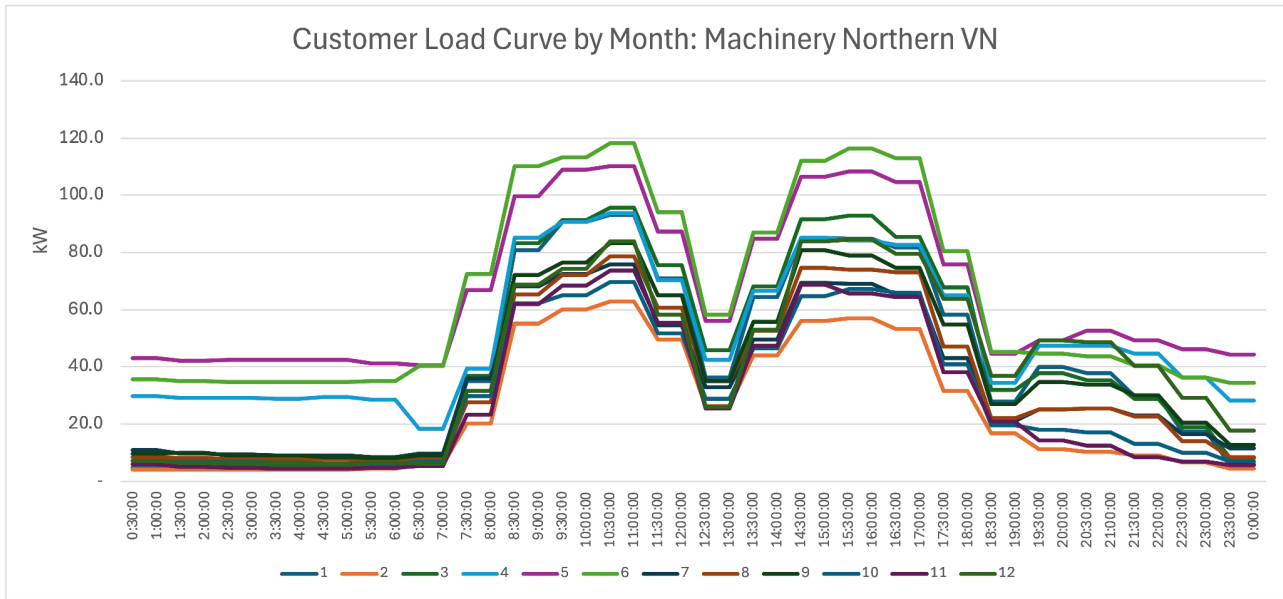


Figure 27. Monthly Machinery Load Curve from Northern Vietnam

Like for the steel and metallurgy customers, the load curves for this customer type show a sharp drop in power demand during the midday hours, with two clear peaks in demand. According to this user's load profile, the BESS system activates in the morning hours as well as in the evening hours, which helps reduce overall system load during those two time periods. However, as can be seen below, the fact that demand during the midday period is low, the best dimensioning of the solar PV system results in a relatively small RTS installation when compared to the overall daily demand. This relative under-dimensioning of the solar system helps avoid spillage, or curtailment.

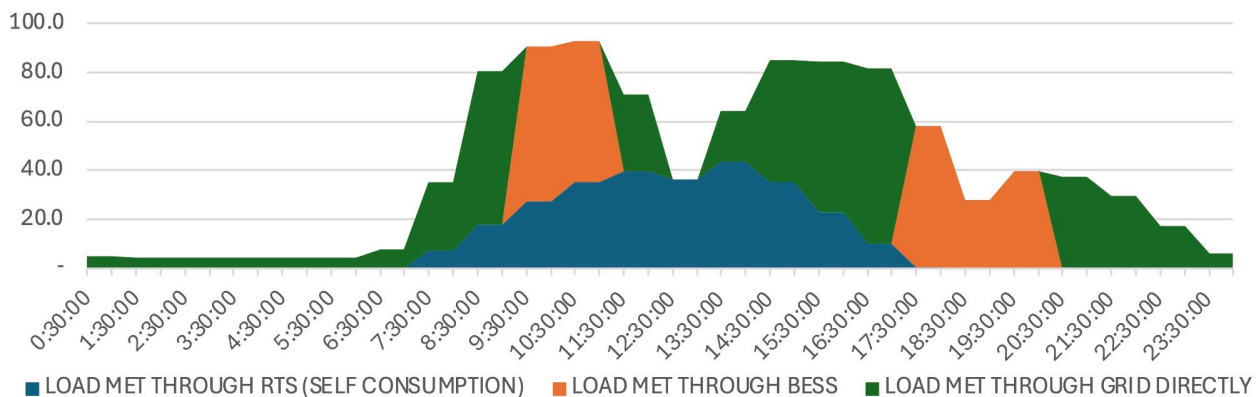


Figure 28. Daily profile of RTS+BESS on a typical day of Machinery customers

As can be seen, the RTS system for this user is considerably smaller than it might otherwise be, considering the customer's overall electricity demand: the reason is mainly due to the shape of the load curve, specifically the sharp drop in demand around midday. For this customer type and load curve, having a large or "over-dimensioned" solar PV system would lead to significantly higher spillage, or curtailment. As a result, the PV system has been dimensioned more conservatively to avoid excessive electricity spillage.

The following charts show the impacts of different levels of penetration of RTS+BESS for the machinery and equipment manufacturing sector, including scenarios for 10%, 25%, and 50% of customers adopting RTS+BESS.

10% RTS+BESS Penetration Rate

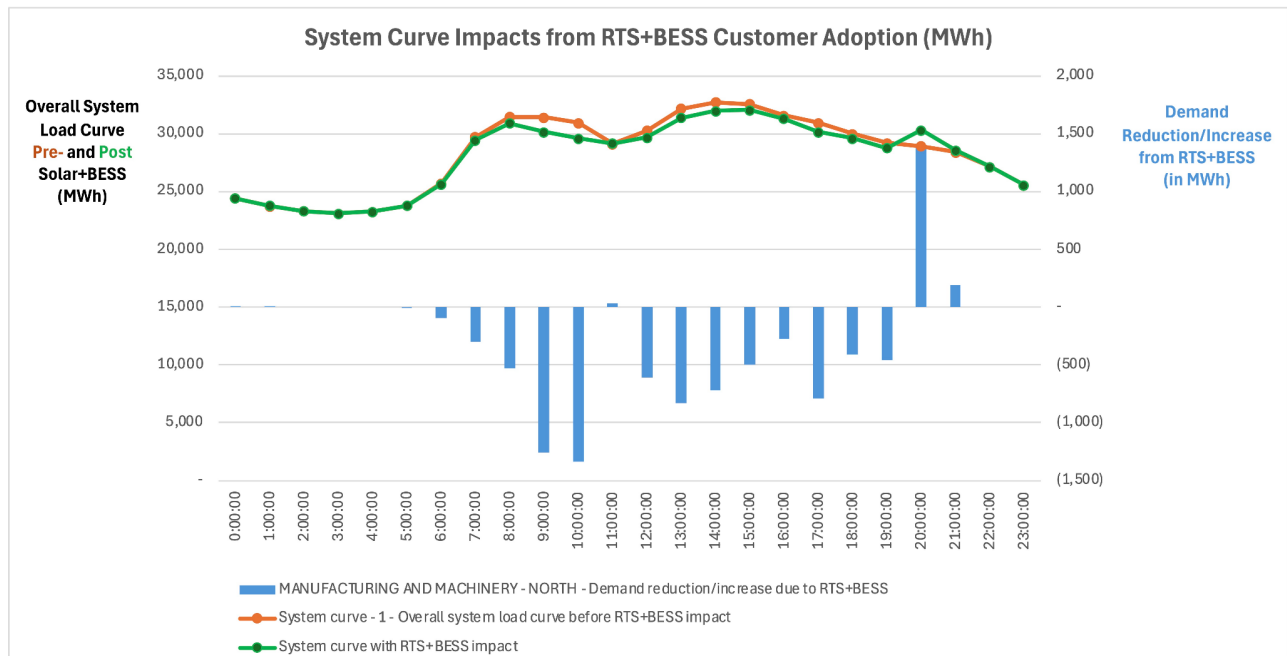


Figure 29. System-level impact of Machinery and Equipment Manufacturing

25% RTS+BESS Penetration Rate:

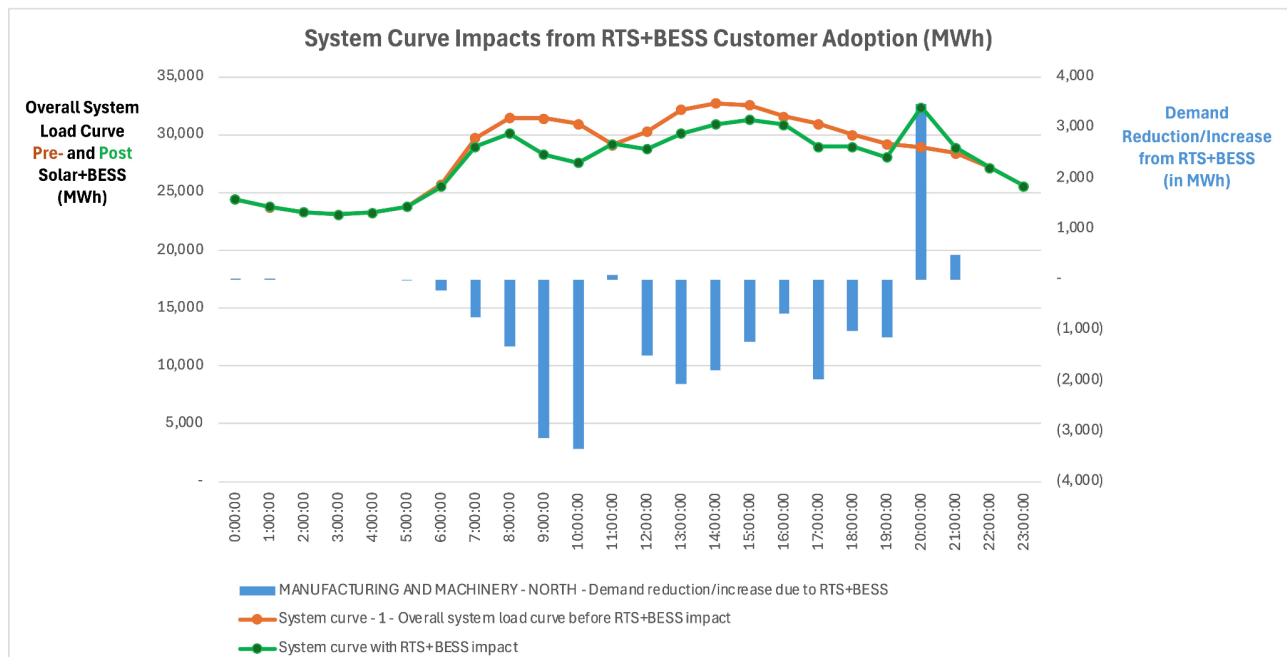


Figure 30. System-level impact of Machinery and Equipment Manufacturing

50% RTS+BESS Penetration Rate:

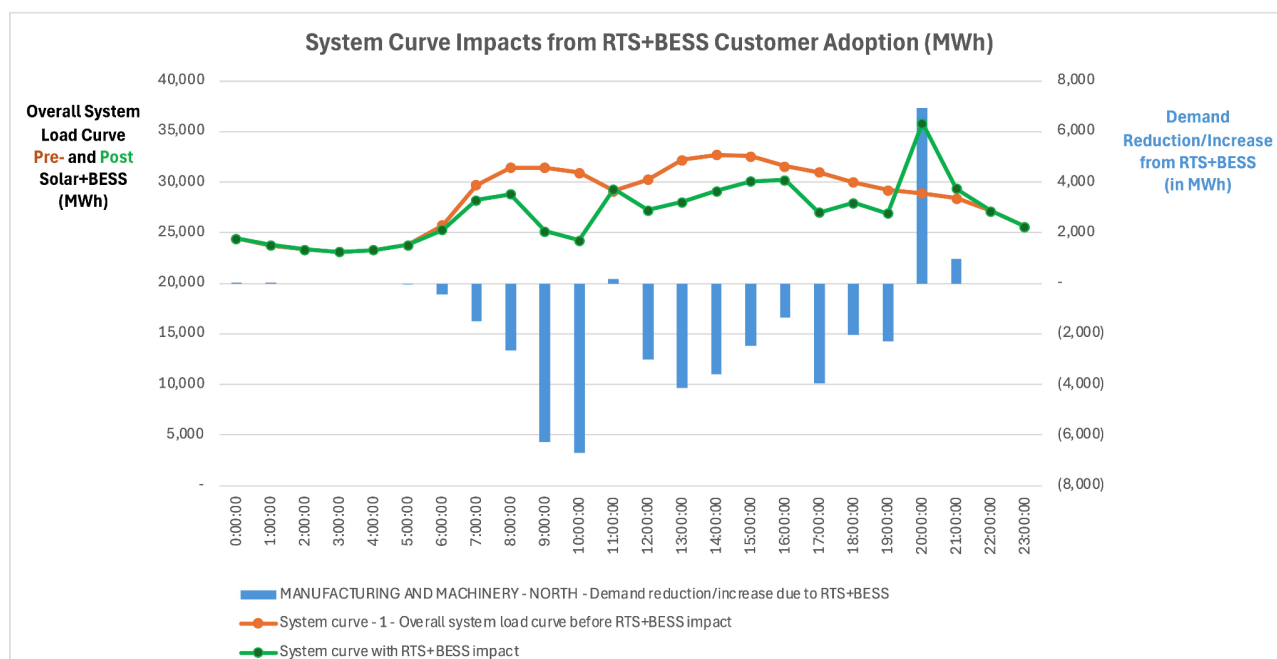


Figure 31. System-level impact of Machinery and Equipment Manufacturing

Key Findings from RTS+BESS adoption in the **Machinery** sector

- **The overall impact on the power system demand curve is significant particularly at higher levels of penetration:** at an estimated 50% RTS+BESS penetration, the aggregate impact on system load is up to **6.700MW** during the daytime.
- In terms of its contribution to mitigating peak electricity demand, the adoption of RTS+BESS in the machinery sector among roughly 50% of the sector's customers would trigger an **11% - 20% reduction** in peak electricity demand.

However, based on Vietnam's current TOU tariff, which declines in the evening, a **clear demand spike can be seen in the evening hours**. At this point, BESS systems are activated and start charging during the "off-peak" tariff hours. Given the current structure and timing of Vietnam's TOU tariffs, this could lead to a spike in power demand at this time. This spike could be mitigated or delayed via regulation, via mandate, or via changes to the TOU tariff structure.

3.5. Aggregate Combined Case

In order to develop a clearer picture of how broad-based RTS+BESS deployment could impact the overall shape of the load curve in Vietnam, it is possible to create scenarios showing different combinations of deployment levels across different customer categories.

	Steel	Hotel	Food Processing	Machinery
Scenario 1: Moderate				
Percentage Adoption Rate	5%	50%	30%	30%
(# of customers adopting)	(15)	(3,472)	(24,499)	(16,200)
Scenario 2: High				
Percentage Adoption Rate	25%	75%	75%	50%
(# of customers adopting)	(73)	(5,208)	(40,500)	(40,832)

Table 6. Summary of main modeling scenarios

Under the moderate scenario 1, the overall impact on the load curve is shown below:

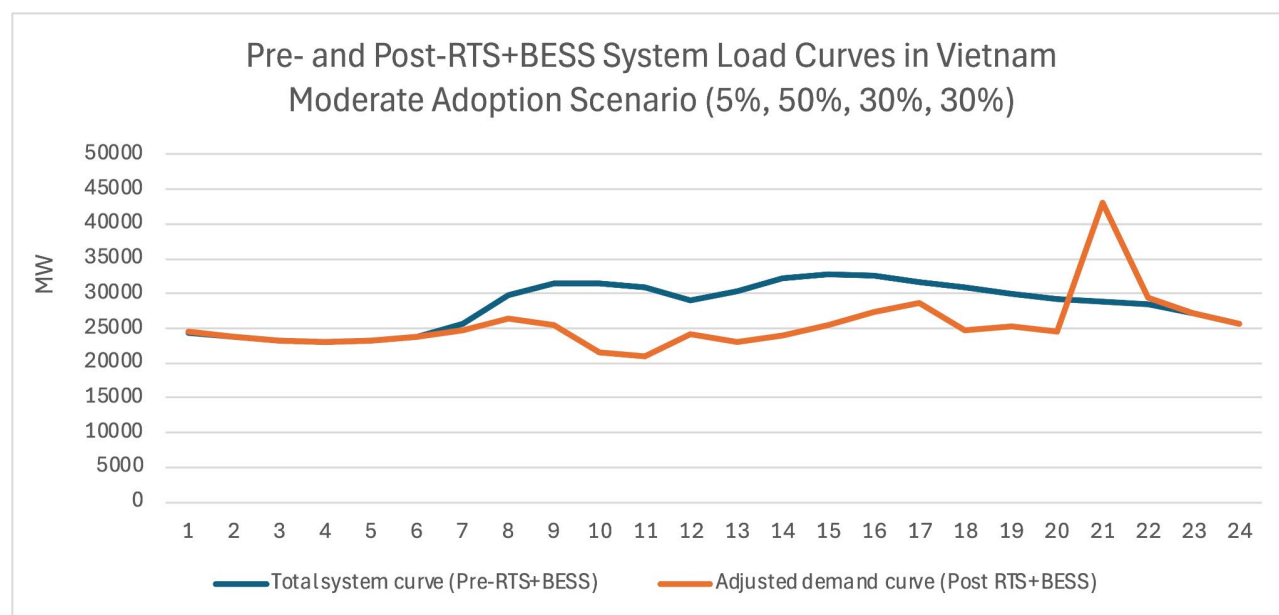


Figure 32. Moderate Adoption Scenario: Illustrative

It is important to note, however, that this load impact only captures the impact of these four sub-sectors (steel, hotel, food processing, and machinery). If additional sub-sectors were added, such as cement, chemical, textile and others, the aggregate impact could be much more substantial.

One recurring feature of these scenarios is that they all imply a sudden spike in electricity demand in the evening hours when BESS systems across the country are activated and start recharging (right after “peak hour at 8:00PM” ends).

In order to give a sense of what such as a “higher” adoption scenario could look like, the chart below shows Scenario 2 with a far larger number of customers adopting RTS+BESS:

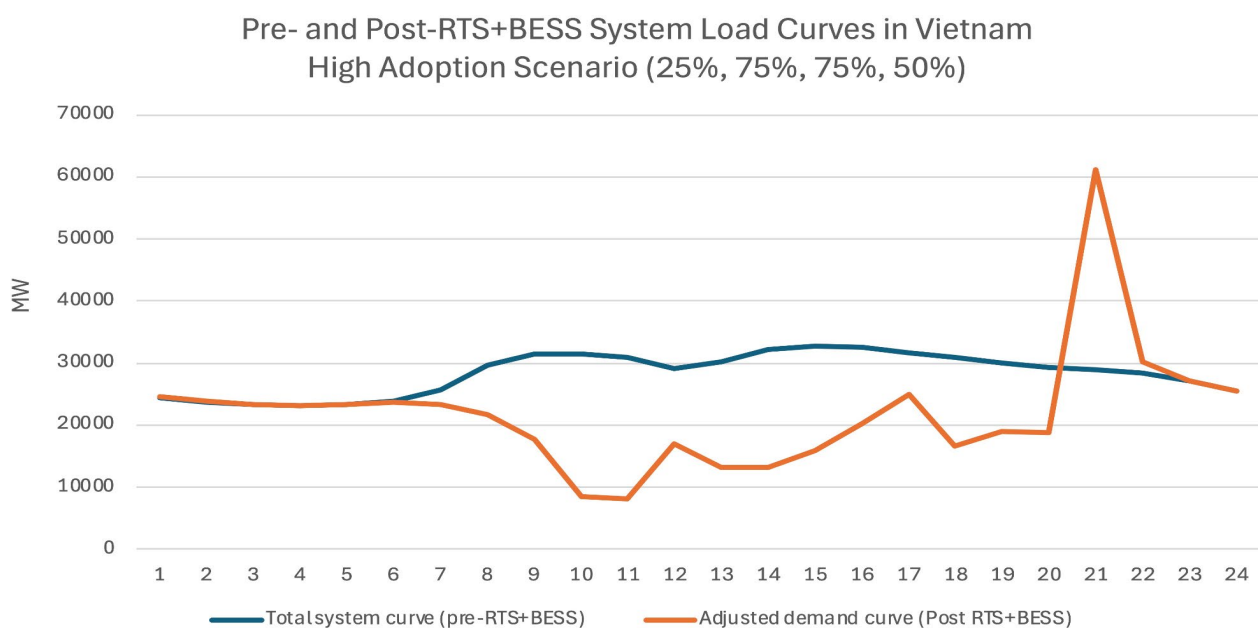


Figure 33. High Adoption Scenario: Illustrative

As can be seen above, the impact of widespread RTS plus BESS adoption can have a significant impact on the overall shape of the load curve. The impact of solar during the daytime in particular can reduce electricity demand substantially: under the scenario shown, the daytime demand curve would drop to below 10GW, before shooting up again the evening as the batteries are charged.

In addition to showing the impacts on the daytime load curve, the last two charts make clear that controlling **the timing of BESS charging becomes increasingly critical as the total quantity of BESS grows**. It would be inefficient for the power system as a whole to have to meet a demand spike caused by BESS, particularly a spike as significant and short-lived as the spikes shown above. Instead of fixing a problem (namely, reducing system peaks), battery charging under TOU tariffs risks actually creating a new problem if all BESS owners configure their assets to start charging at the same time.

This underscores why shifting or staggering the TOU tariff timing could be a valuable policy intervention to consider, as it could help smooth the overall impact on the system, spreading it out through the evening “off-peak” hours and thereby help smooth out the overall shape of the demand curve.

4. Financial Analysis of RTS plus BESS

The cases differ widely depending on the customer load curve, system dimensioning, location in Vietnam (North vs. South), as well as depending on the specific applicable tariff category applied. To avoid unnecessary complexity, the assumed voltage level for all customers was between 22kV – 110kV.

As part of the modelling, it is possible to model the load curves under one electricity tariff, or the other, as well as at different voltage levels. Since the tariffs for commercial customers are significantly higher than the tariffs applied to industrial customers, even for customers connected at the same voltage level, customers operating under Vietnam’s “commercial” electricity tariffs show a significantly higher profitability (IRR) than if they were operating under the lower “industrial” customer tariffs.

The table of tariffs used is shown below, based on MOIT's Decision No.1279/QĐ-BCT dated 09 May 2025 on electricity prices:

	Industry Tariffs (22kV – 110kV)	Commerce Tariffs (22kV – 110kV)	% Difference (Commerce vs Industry tariffs)
Regular hours	1.833 VND/kWh	2.887 VND/kWh	58% higher
Off-peak	1.190 VND/kWh	1.609 VND/kWh	35% higher
On-peak	3.398 VND/kWh	5.025 VND/kWh	48% higher

Table 7. Overview of Commercial and Industrial Electricity Tariffs in Vietnam (22kV – 110kV)

Given the significantly higher tariffs for commercial customers, the profitability for customers under these tariffs is markedly higher.

Summary of financial results:

Although it is impossible to summarize the profitability for every individual customer, and every individual dimensioning of RTS and BESS systems, by running multiple scenarios and sensitivities, it is possible to summarize the broad range of profitability of investing in RTS plus BESS in Vietnam under different tariff categories. It is also possible to model the profitability for customers with significantly different load curves. For instance, some of the EVN load curves provided had highly unusual demand profiles, with a significant amount of demand occurring only in the evening between 10PM and 4AM. For such customers, adding RTS plus BESS makes far less economic sense, and IRRs were often in the low single digits (3-7%). For customers with a high daytime demand, such as food processing and hotels, the financial attractiveness is on average far higher, with some customer load curves showing IRRs above 30% (with payback times of approximately 3 years).

Another factor that is important to consider is the financing structure of the project, specifically whether the project uses any debt, and if so, how much. The table below provides a summary of the impact of different levels of debt (0%, 50%, and 70%) in the overall financing structure, assuming the cost of debt of 6.23%.

	0% debt	50% debt	70% debt
Steel Customer (North)	13.3%	14.5%	15.2%
Food Processing (South)	15.5%	17.8%	19.0%
Hotel (South)	24.7%	30.7%	34.4%
Machinery (North)	12.7%	14.1%	14.9%

Table 8: Impact of capital structure on project returns (IRRs)

In order to show the range of profitability under different tariff levels, the table below shows a summary of the IRR and payback time under both tariff categories (commerce & industry tariffs):

Consolidated Financial Returns (combining both RTS and BESS)		
	Customer under Industry Tariffs (22kV - 110kV)	Customer under Commerce Tariffs (22kV - 110kV)
Steel	IRR: 13.3% Payback time: 8 years	N/A
Food Processing	IRR: 15.5% Payback time: 7 years	N/A
Hotel²²	IRR: 15.1% Payback time: 7 years	IRR: 24.7% Payback time: under 5 years
Machinery	IRR: 12.7% Payback time: 9 years	N/A

Table 9. Overview of consolidated financial returns for RTS and BESS Overview of consolidated financial returns for RTS and BESS

²² Hotel sub-sector will be soon applies Industrial tariffs follows Decision 14/2025/QĐ-TTg dated 29th May 2025

Overall, IRRs for customers under commercial tariffs are roughly 9% higher than for customers under the industrial tariffs. This means that in terms of the future evolution of the market in Vietnam, with the current TOU tariffs, **it is more likely for a sustained increase in RTS plus BESS to occur among customers under commercial electricity tariffs.**

One key finding that is consistent across all cases is that the actual financial profitability stems from the RTS investment, and not from the BESS investment as such.

In other words, **when the financials of both RTS and BESS are considered separately (i.e., on a non-consolidated basis), the profitability of a well-dimensioned RTS system (with little to no spillage) at today's solar PV prices (of approximately VND 10.5 million/kW) is very attractive, and is ultimately what drives the attractiveness of the investment.** When looking strictly at the profitability of the RTS investment, the IRRs for a well-dimensioned solar PV system for a customer under commercial tariffs are frequently above 40% with payback times of under 3 years. For a customer under industrial tariffs, the IRRs purely for the RTS investment typically ranged from 25%-33%.

By contrast, when a BESS system is added, **the BESS asset (considered on a stand-alone basis) weighs down the profitability of the overall investment.** When BESS is added, the profitability on a consolidated basis for most cases drops from (for instance) 30% to 15% for customers under industry tariffs, and from 45% down to approximately 25% for customers under commercial tariffs. Note that this assumes customers with a relatively high daytime power demand that coincides well with solar PV output.

Considered on a stand-alone basis, the IRR for larger BESS systems remains marginal or even negative for many customer load profiles (see Tables 10 - 15). However, when dimensioned at smaller sizes, some BESS investments demonstrate modest returns, even on a stand-alone basis (without RTS). **When combined with RTS, many BESS projects demonstrate attractive returns on a consolidated basis.** The tables below show a range of sensitivity analyses for all four customer types, covering a full range of RTS and BESS system sizes.

Food Processing Customer (Ho Chi Minh City)		BESS System Size (kWh)								
RTS Size (kW)		175	350	525	700	875	1,050	1,225	1,400	1,575
-	n.m.	8.7%	8.7%	8.6%	6.9%	2.5%	(1.5%)	(6.0%)	(11.9%)	n.m.
175	32.2%	25.0%	21.2%	17.9%	14.9%	11.7%	8.9%	6.2%	3.7%	1.3%
350	32.0%	27.5%	23.6%	20.7%	18.1%	15.3%	12.9%	10.7%	8.8%	7.0%
525	28.1%	24.7%	22.1%	19.9%	17.8%	15.5%	13.5%	11.7%	10.0%	8.5%
700	22.9%	20.6%	18.8%	17.3%	15.7%	13.8%	12.1%	10.6%	9.2%	7.8%
875	18.8%	17.3%	16.0%	14.8%	13.5%	12.0%	10.5%	9.2%	7.9%	6.7%
1,050	15.8%	14.6%	13.7%	12.8%	11.7%	10.3%	9.1%	7.8%	6.7%	5.5%
1,225	13.3%	12.4%	11.7%	10.9%	10.0%	8.8%	7.6%	6.5%	5.4%	4.4%
1,400	11.2%	10.5%	9.9%	9.3%	8.5%	7.4%	6.3%	5.3%	4.2%	3.2%
1,575	9.5%	8.9%	8.4%	7.9%	7.1%	6.1%	5.1%	4.1%	3.1%	2.1%

Table 10. Sensitivity Analysis (1)

Financial attractiveness (IRR) of RTS, BESS, and RTS+BESS for a Food Processing Customer in Southern Vietnam (under Industrial tariffs, 22kV – 110kV)²³

Manufacturing/Machinery Customer (Hanoi)		BESS System Size (kWh)								
RTS Size (kW)		88	175	263	350	438	525	613	700	788
-	n.m.	8.7%	8.5%	7.3%	5.4%	2.0%	(1.7%)	(6.2%)	(12.3%)	n.m.
88	20.4%	16.9%	14.9%	13.0%	10.5%	7.7%	5.0%	2.1%	(0.8%)	(3.8%)
175	20.1%	18.2%	16.6%	14.5%	12.3%	9.9%	7.6%	5.4%	3.3%	1.3%
263	19.3%	18.1%	16.5%	14.7%	12.7%	10.6%	8.7%	6.9%	5.2%	3.5%
350	18.0%	17.1%	15.6%	14.0%	12.4%	10.6%	8.9%	7.4%	5.9%	4.4%
438	16.3%	15.3%	14.2%	12.8%	11.4%	9.8%	8.4%	7.0%	5.7%	4.4%
525	14.4%	13.5%	12.5%	11.4%	10.1%	8.8%	7.5%	6.3%	5.1%	3.9%
613	12.5%	11.8%	10.9%	10.0%	8.9%	7.7%	6.5%	5.3%	4.2%	3.1%
700	10.8%	10.2%	9.5%	8.6%	7.6%	6.5%	5.4%	4.4%	3.3%	2.3%
788	9.2%	8.7%	8.0%	7.3%	6.4%	5.3%	4.3%	3.3%	2.3%	1.3%

Table 11. Sensitivity Analysis (2)

Financial attractiveness (IRR) of RTS, BESS, and RTS+BESS for a Machinery Customer in Northern Vietnam (under Industrial tariffs, 22kV – 110kV)

Machinery Customer (Southern Vietnam: HCMC)		BESS System Size (kWh)								
RTS Size (kW)		38	75	113	150	188	225	263	300	338
-	n.m.	8.7%	7.5%	5.2%	3.1%	1.4%	(0.0%)	(1.6%)	(3.8%)	(8.0%)
38	32.2%	25.0%	20.9%	17.6%	14.9%	12.7%	10.7%	8.6%	6.2%	4.0%
75	32.2%	27.9%	24.7%	21.8%	19.2%	17.0%	14.6%	12.4%	10.5%	8.7%
113	32.1%	28.9%	26.3%	23.7%	21.2%	18.8%	16.6%	14.7%	13.0%	11.4%
150	31.5%	29.1%	26.8%	24.2%	21.8%	19.6%	17.8%	16.1%	14.6%	13.2%
188	30.3%	28.2%	26.0%	23.7%	21.6%	19.8%	18.1%	16.6%	15.3%	14.0%
225	27.9%	26.1%	24.3%	22.4%	20.6%	19.0%	17.6%	16.3%	15.0%	13.9%
263	25.2%	23.7%	22.2%	20.6%	19.1%	17.8%	16.5%	15.3%	14.3%	13.2%
300	22.7%	21.5%	20.3%	18.9%	17.6%	16.4%	15.3%	14.3%	13.3%	12.3%
338	20.4%	19.4%	18.4%	17.2%	16.1%	15.0%	14.0%	13.1%	12.2%	11.3%

Table 12. Sensitivity Analysis (3)

Financial attractiveness (IRR) of RTS, BESS, and RTS+BESS for a Machinery Customer in Southern Vietnam (under Industrial tariffs, 22kV – 110kV)

²³ Note that if food processing customers were to fall under the « industrial tariff » category, the returns would be correspondingly lower, as the industrial tariff is lower.

Steel Customer (Bac Ninh, Hanoi Region)		BESS System Size (kWh)								
RTS Size (kW)	-	28,438	56,875	85,313	113,750	142,188	170,625	199,063	227,500	255,938
-	n.m.	8.7%	8.5%	7.3%	4.8%	1.5%	(1.8%)	(6.0%)	(12.0%)	n.m.
20,313	20.4%	16.0%	14.0%	12.0%	9.3%	6.5%	3.8%	0.6%	(2.7%)	(6.3%)
40,625	20.4%	17.7%	15.9%	14.0%	11.4%	9.0%	6.4%	3.9%	1.5%	(1.0%)
60,938	20.4%	18.4%	16.9%	15.0%	12.6%	10.4%	8.1%	6.0%	4.0%	2.0%
81,250	20.1%	18.7%	17.2%	15.4%	13.2%	11.1%	9.1%	7.2%	5.5%	3.8%
101,563	19.6%	18.4%	17.0%	15.2%	13.3%	11.3%	9.5%	7.9%	6.3%	4.9%
121,875	18.8%	17.8%	16.4%	14.8%	13.0%	11.2%	9.7%	8.2%	6.8%	5.4%
142,188	17.9%	16.9%	15.6%	14.2%	12.5%	11.0%	9.5%	8.2%	6.9%	5.6%
162,500	16.7%	15.8%	14.6%	13.3%	11.8%	10.4%	9.1%	7.9%	6.7%	5.5%
182,813	15.4%	14.5%	13.5%	12.3%	11.0%	9.7%	8.5%	7.3%	6.2%	5.1%

Table 13. Sensitivity Analysis (4)

Financial attractiveness (IRR) of RTS, BESS, and RTS+BESS for a Steel Customer in Northern Vietnam (under Industrial tariffs, 22kV – 110kV)

Hotel Customer (Southern Vietnam, HCMC)			BESS System Size (kWh)							
RTS Size (kW)	-	138	275	413	550	688	825	963	1,100	1,238
-	n.m.	15.1%	15.1%	14.9%	12.4%	8.3%	4.7%	1.6%	(1.4%)	(4.6%)
138	49.5%	38.1%	31.9%	26.9%	23.4%	19.7%	16.6%	14.1%	11.7%	9.6%
275	48.8%	41.0%	35.2%	31.2%	28.1%	24.7%	21.8%	19.3%	17.1%	15.2%
413	38.6%	34.0%	30.5%	27.9%	25.7%	23.1%	20.7%	18.6%	16.8%	15.1%
550	30.6%	27.9%	25.7%	23.9%	22.4%	20.4%	18.4%	16.7%	15.2%	13.8%
688	25.2%	23.4%	22.0%	20.7%	19.5%	17.9%	16.3%	14.9%	13.6%	12.3%
825	21.3%	20.1%	19.0%	18.1%	17.2%	15.8%	14.5%	13.2%	12.0%	11.0%
963	18.3%	17.4%	16.6%	15.9%	15.1%	14.0%	12.8%	11.7%	10.6%	9.6%
1,100	15.9%	15.2%	14.6%	14.0%	13.4%	12.4%	11.3%	10.3%	9.4%	8.5%
1,238	13.9%	13.4%	12.9%	12.5%	11.9%	11.0%	10.0%	9.1%	8.2%	7.4%

Table 14. Sensitivity Analysis (5)

Financial attractiveness (IRR) of RTS, BESS, and RTS+BESS for a Hotel Customer in Southern Vietnam (under Commercial tariffs, 22kV – 110kV)

Hotel Customer (Bac Ninh, Hanoi Region)		BESS System Size (kWh)								
RTS Size (kW)	-	175	350	525	700	875	1,050	1,225	1,400	1,575
-	n.m.	15.1%	15.1%	13.3%	9.7%	5.8%	2.1%	(1.4%)	(5.0%)	(9.8%)
175	31.9%	26.5%	23.6%	20.2%	16.6%	13.4%	10.7%	8.2%	5.8%	3.6%
350	31.9%	28.6%	25.7%	22.6%	19.6%	16.8%	14.5%	12.3%	10.3%	8.5%
525	31.2%	28.6%	25.8%	23.3%	20.8%	18.5%	16.3%	14.4%	12.7%	11.2%
700	28.1%	25.9%	23.9%	22.0%	20.0%	17.9%	16.1%	14.5%	13.0%	11.6%
875	24.3%	22.6%	21.2%	19.8%	18.1%	16.4%	14.8%	13.4%	12.1%	10.9%
1,050	21.0%	19.8%	18.7%	17.6%	16.2%	14.7%	13.4%	12.2%	11.0%	9.9%
1,225	18.2%	17.3%	16.5%	15.6%	14.4%	13.1%	12.0%	10.9%	9.8%	8.8%
1,400	15.9%	15.2%	14.6%	13.8%	12.7%	11.6%	10.6%	9.6%	8.7%	7.7%
1,575	13.9%	13.4%	12.9%	12.3%	11.3%	10.3%	9.3%	8.4%	7.5%	6.7%

Table 15. Sensitivity Analysis (6)

Financial attractiveness (IRR) of RTS, BESS, and RTS+BESS for a Hotel Customer in Northern Vietnam (under Commercial tariffs, 22kV – 110kV)

The implications of this are significant: **under the 2025 TOU tariffs in Vietnam, adding BESS to an RTS investment to fully displace peak tariffs is not (yet) financially attractive.** The BESS ends up reducing the overall (consolidated) IRR of the investment for most customer types. In many of the cases, the profitability of the BESS investment on a stand-alone basis is zero, or only very marginally profitable, under current TOU tariffs.

However, as can be seen, in many cases, adding a modest (under-dimensioned) BESS system can maintain good financial performance while potentially providing additional benefits to the user in terms of RTS self-consumption, environmental performance, and electricity reliability. Such **smaller, more modest battery sizes can be considered financially attractive**, particularly when combined with RTS.

In order to evaluate the impacts of potential changes to the TOU structure, it is possible to model the profitability under different TOU rates. One of the main ways in which the attractiveness of BESS can be increased is by increasing the spread between “non-peak” and “on-peak” tariffs: this makes it more attractive for customers to store electricity during the daytime via RTS, or via the grid overnight, and to activate the BESS to avoid these high spikes. A further option would be to lengthen the peak tariff period – however, a longer peak tariff period would require a larger battery, and may be opposed by certain industry groups, which may make it less attractive than an increase to the TOU rate itself.

The table below shows the impact on profitability for both hotel and food processing customers in Vietnam under the current TOU tariffs and under the hypothetical introduction of a TOU tariff with a far higher “**super peak**” tariff (or “critical peak pricing”) (modelled at VND 10.000/kWh, or EUR 0.32/kWh, which represents nearly double the current peak tariff level) to illustrate the impact of higher TOU rates. This “super peak” tariff is presented for illustrative purposes, and reflects some of the TOU tariff changes that other jurisdictions are introducing elsewhere in the world, as seen in the case studies previously. Higher TOU rates create a greater incentive to mitigate peaks, and can help catalyze broader investments in BESS. However, it is important to note that such “**super peak**” tariff options can have negative impacts on customers without BESS, or those without the ability to adjust their onsite demand to avoid peak prices; this is one reason why “super peak” tariffs are typically structured on an opt-in basis (i.e., Critical Peak Pricing – CPP for Demand Response programme).²⁴ As highlighted previously, the current profitability of stand-alone BESS investments in Vietnam is marginal, or even slightly negative. In the cases shown here, the “stand-alone” BESS investments under current TOU rates even show negative IRRs: in other words, in these cases, it is only by adding solar into a combined project that the consolidated returns become positive.

²⁴ Synapse (2025).

<https://www.synapse-energy.com/sites/default/files/22-086%20Comments%20on%20TOU%20Rates.pdf>

Scenarios	Hotel (Northern VN) ²⁵	Hotel (Southern VN) ²⁶	Food Processing (Southern VN) ²⁴
Current TOU rates: RTS only	31.2%	48.8%	43.3%
Current TOU rates: BESS only	-1.1%	-1.7%	-6.7%
Current TOU rates: Consolidated (RTS+BESS)	20.8%	24.7%	25.5%
"Super peak" TOU scenario: RTS only	41.3%	64.4%	56.8%
"Super peak" TOU scenario: BESS only	25.2%	24.3%	19.9%
"Super peak" TOU scenario: Consolidated (RTS+BESS)	34.8%	41.9%	40.0%

Table 16. Financial attractiveness (IRR) of RTS, BESS, and RTS+BESS investments under "current" and "super peak" tariff scenarios

However, as can be seen above, the profitability under the "**super peak**" simulation (assuming with TOU tariffs of VND 10,000/kWh) shows significantly higher IRRs and helps make the BESS investment profitable. Since the BESS investments have a lower IRR than the stand-alone RTS investment, the impact overall still ends up reducing the consolidated IRR: however, the consolidated investment remains financially attractive as a whole.

²⁵ Reflec to actual solar irradiance/PV output in Northern Region.

²⁶ Reflec to actual solar irradiance/PV output in Sourthern Region

5. Conclusion: Overview of TOU Rate Options for Vietnam

Overall, RTS plus BESS systems in Vietnam are becoming increasingly financially attractive. Returns differ depending on the unique location of the customer, the shape of the load curve, the specific tariff category (commercial or industrial), as well as the specific dimensioning of the RTS and/or BESS system. Based on the single-part/energy-only electricity tariff in place in mid-2025 according to Decision No.1279/QĐ-BCT dated 09 May 2025 on electricity prices, the returns are sufficiently attractive for smaller BESS system sizes to be attractive, particularly when combined with RTS.

Nevertheless, given that electricity reliability remains relatively high in Vietnam, many customers may prefer to simply adopt RTS and wait before adopting BESS. RTS offers attractive returns on a stand-alone basis across all customer types, and all regions of Vietnam. The combined attractiveness of RTS plus BESS is likely to improve further as **the two-part electricity tariff structure** is introduced, as customers can benefit further by reducing their fixed demand charges, but further analysis is needed to evaluate the impacts.

In addition, the impact on Vietnam's overall load curve of widespread RTS plus BESS uptake can be considerable, depending on the level of adoption. This report has shown different levels of adoption based on a more moderate adoption level, as well as a "high" adoption scenario, and **this report has found that the potential of RTS and BESS to reshape Vietnam's overall system load curve (and reduce electricity demand peaks) is significant.**

As seen in the previous section on TOU rates, there are many different ways of structuring TOU rates. Given Vietnam's context and current TOU rates, the recommendations presented here are divided into two parts:

1. **Recommendations to change the overall shape of the load curve** and reduce current demand peaks.
2. **Recommendations to adjust the specific time at which customers with BESS choose to start recharging their BESS systems.**

5.1. Recommendations to change the overall shape of the load curve

The modeling conducted for this analysis is based on the electricity tariff structure in place in Vietnam until mid-2025 according to Decision No.1279/QĐ-BCT dated 09 May 2025. **It does not include the new "two-part" electricity tariff**, which features both a capacity component, and an energy component. In general, having a capacity component in the electricity tariff structure is advantageous from a system perspective, as it provides for a better alignment between electricity rates and overall system costs. With the same total energy consumption, the

customers who can reduce their maximum power demand (P_{max}) will pay less overall, while also helping the electricity system reduce peak load. The shift toward a two-part electricity structure (energy and capacity) should contribute to more efficient system operation, lower investment costs, and better overall cost recovery and operational efficiency.

Introducing a fixed capacity payment also tends to improve the attractiveness of BESS investments: many commercial and industrial customers that have adopted BESS in the last 5–8 years in the US for instance have done so primarily to avoid high (and rising) capacity charges. BESS systems can be configured to help customers stay below certain capacity thresholds, and thereby provide real and immediate cost savings.

1	Expand the TOU analysis to include the new two-part electricity tariff structure. Expanding the analysis of the impact of Solar plus BESS adoption to include the latest electricity tariff structure can provide a more up-to-date and accurate picture of how the two-part electricity structure, as being implemented in Vietnam, will impact the financial attractiveness of solar plus BESS investments.
2	Consider lengthening the duration of the morning peak period: Given Vietnam’s overall power demand curve, MOIT may wish to consider extending (i.e. lengthening) the peak tariff period to cover more daytime hours. A longer peak tariff (e.g., from 9:30 to 14:30) would create a clearer incentive for RTS investments in particular. A longer peak period during the daytime would send a clear signal to invest in solar PV for commercial and industrial customers, and could help flatten, or even <i>invert</i>, the daytime load curve of the power system.
3	Consider increasing the TOU “peak tariff” rate or even creating a “super peak” tariff, similar to other jurisdictions such as the US States of California and Maryland: As seen above, increasing the peak tariff makes both BESS and the consolidated RTS+BESS investments much more attractive in Vietnam. In order to help reduce high demand peaks in the overall power system, such as in summer evenings due to growing air conditioner use, MOIT could consider introducing a “super peak” tariff during the summertime for commercial and industrial customers, or as part of the demand response (DR) program. This would help encourage such customers to add RTS plus BESS, and thereby contribute to reducing system peaks. Such a critical peak price could be introduced as part of Vietnam’s upcoming demand response program and can be designed as an “opt-in” tariff option. Customers that have the ability to add BESS or to be more flexible can therefore choose to opt-into the “super peak” tariff. In response, the normal and off-peak rates generally have to be correspondingly lower than the normal TOU rate to make such super peak tariffs attractive.

4

Consider the introduction of “peak time rebates” instead of higher TOU rates: peak time rebates offer customers a credit on their bill for avoiding demand during critical peak periods. Experience in jurisdictions such as the US shows that such “rebates” generate less pushback from industry and can still help achieve some valuable peak shaving benefits. Obtaining such a “peak time rebate” would also create an additional “revenue stream” for customers installing BESS, which may help improve the economics, and avoid needing to introduce higher TOU peak rates.

5.2. Recommendations to avoid large-scale simultaneous charging behavior

The recommendations provided in this subsection are designed specifically to discourage simultaneous recharging. Under Vietnam’s electricity tariff structure, all customers within a given category are subjected to the same pricing structure: this increases the risks of sudden spikes in demand due to simultaneous BESS recharging, as BESS systems are often configured to switch on once lower tariffs apply. While this issue may seem hypothetical today, the impacts of simultaneous charging on the overall system load curve can be significant as BESS adoption grows, both among commercial and industrial customers, as well as among electric vehicle chargers (or charging station operators).

1

Consider shifting the “off-peak” tariff to later in the evening for commercial and industrial customers in order to avoid large, BESS-charging related demand spikes: this would encourage more RTS+BESS customers to shift their BESS charging to later in the evening rather than switching on at 8:00PM as is currently the case. The spike in electricity demand from BESS charging can be seen across all modelled cases, and can be expected in real-world conditions, as the algorithms used by BESS installers and operators tend to be designed to match the TOU rate structures in place. This can be expected to lead to sharp, discontinuous “spikes” in demand due to the sudden activation of BESS assets as they enter charging mode.

2

Consider staggering the timing of TOU rates by customer class to avoid widespread simultaneous charging: rather than having all industrial customers subjected to the same TOU rates, with the same time-based structure, Vietnam could consider adjusting the timing of the TOU rates by customer class. In markets with competitive wholesale markets, customers connected at higher voltages (such as over 110kV) are occasionally permitted (or required) to purchase their electricity at wholesale market prices. This direct wholesale market price exposure effectively achieves the same objective: if market prices already start dropping at around 8:15PM or 8:30PM, the customers exposed to market prices can start to adapt their consumption accordingly; similarly, if wholesale market prices remain higher until

	9:30PM, they can hold off, and avoid large surges in demand until prices come back down. In Vietnam's case, MOIT may wish to consider introducing new electricity tariff categories in the future. This could mean creating new tariff sub-categories with the same tariff levels overall, but with slightly different timing (e.g. normal tariffs starting at 9PM or 9:30PM instead of 8PM, and the "off-peak" tariff starting later in the evening.) Such a change could be applied by introducing a different "time structure" between commercial and industrial customer categories, for instance. Note that while such a change would also automatically have an impact the overall load curve, the main policy objective of such a change is to "stagger" or shift the time at which customers with BESS recharge their BESS systems.
3	Consider staggering the timing of TOU rates by voltage level to avoid large-scale simultaneous charging behavior: instead, the TOU tariff could be differentiated with a slightly different structure for customers connected at different voltage levels (e.g., between 22kV - 110kV and greater than 110kV). In other words, the customers connected to 22kV - 110kV could be given a shorter (or longer) duration under the normal tariff and earlier (or later) access to "off-peak" pricing, whereas customers above 110kV could be given later access to off-peak pricing in order to stagger the BESS charging "start times".
4	To avoid further simultaneous charging from EVs, consider introducing a new tariff category specifically for EV chargers, and/or EV charging stations. Such EV-specific charging rates are currently found in India, in the UK, as well as in some US states and are expected to grow as the share of EVs grows. EV-specific charging rates tend to be more closely aligned with wholesale market prices, and in many cases are designed to be dynamic (rather than fixed in advance). Such EV-specific charging rates can be implemented on a mandatory basis (for all charging stations or EV chargers, including those located on private homes or businesses) or on an "opt-in" basis.

With further guidance from MOIT, a range of related scenarios and possibilities can be modelled to evaluate the impacts on the system load curve and to help provide input into potential future changes in the TOU tariff structure in Vietnam, including modeling specifically the impacts on the Northern Power Company, the Southern Power Company, or other areas of the system with specific levels of Solar plus BESS uptake. By updating the tariff structure to match the new two-part electricity tariff, it would also be possible to compare the attractiveness of RTS plus BESS under the new tariff regime being introduced.

